CHEMISTRY Department
SIMHADRI

**Note No :1**

**Date:** 10/07/2025(10:38:17)

**Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Subject:**LMI REVISION FOR CHEMISTRY STARTUP SHUTDOWN LAY UP PRACTICES



OGN FOR CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES has been Revised by CC-OS.

| Previous LMI Version | Present LMI Vesrion |
|---|---|
| COS-ISO-00-OGN/OPS/CHEM/017 Rev. No.: 2 Feb. 2023 | COS-ISO-00-OGN/OPS/SYST/065 ; Rev. No.: 0 |

Changes in LMI are briefed in the below.

1. Monitoring of Condensate ACC/DCC for boiler filling and light-up clearance has been incorporated in drum-type units.

2. A new clause has been added for Hot well-deaerator closed circulation.

3. Monitoring of the main steam CC for turbine rolling clearance has been added.

4. CBD operation guidelines have been updated.
5. Guidelines for installing an automatic pressure-reducing system for SWAS samples in units with low sample flow during cyclic loading have been added.

6. Additional PI tags for subcritical and supercritical units.

7. Condensate dumping pipeline dimensions for 110/210/250/500/600/800MW units have been added in

Annexure.

Draft LMI & OGN are attached for refetrence.

Put up for Kind approval of CA for implementation.

**Submitted by:**


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**On:**10/07/2025(10:38:17) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:2**



Draft LMI is initiated. Copy of Draft LMI and OGN is attached.

All departments are requested to go through the document and comment if any.

Submitted for approval please.

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**On:**10/07/2025(10:49:48) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:3**

Please review.

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**Note No:4**

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

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**Note No:5**



Necessary modification done as per the guidelineds mentioned in LMI/OGN/GEN/002.

put up for the approval please.

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**Note No:6**



Header on every page should mention rev. no. along with LMI No.

**Submitted by:**

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**Note No:7**



Necessary changes incorporated.

submitted for approval.

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**Note No:8**



reviewed

**Submitted by:**

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**On:**22/07/2025(18:19:29) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:9**

LMI is in line with the revised OGN.

May please be approved.

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**Note No:10**

Checked and forwarded.

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**On:**01/08/2025(13:56:48) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:11**



LMI revised version is in line with the OGN document. May please be processed further.

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**Note No:12**



May kindly review

**Submitted by:**

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**Note No:13**

LMI is in line with the revised OGN.

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**Note No:14**

**Submitted by:**

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**On:**04/08/2025(10:01:35) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:15**

Please review.

**Submitted by:**

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**On:**04/08/2025(11:27:10) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:16**



Reviewed C&I points in line wih attched OGN.

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**On:**05/08/2025(19:23:57) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:17**



LMI reviewed.

**Submitted by:**

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**On:**05/08/2025(20:02:51) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:18**



LMI is in line with the OGN. May please be approved.

**Submitted by:**

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**On:**06/08/2025(16:30:06) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:19**



LMI is in line with the OGN.

Submitted for kind approval, please.

**Submitted by:**

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**Note No:20**

**Submitted by:**

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**On:**08/08/2025(18:10:21) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846

**Note No:21**

**Approved & Submitted by:**

समीर शर्मा - कार्यकारी निदेशक

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**On:**08/08/2025(18:59:02) **Ref:**SMTPP/Chemistry/2025-26/CHEM LMI/300846



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**NTPC SIMHADRI**

# LOCATION MANAGEMENT INSTRUCTION

# CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES

**LMI/OGN/OPS/CHEM/017**

**Rev. No. : 04**

**Approved for Implementation by: HOP (SIMHADRI)**

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## LMI APPROVAL DOCUMENT

| | | |
|---|---|---|
| 1 | LMI NAME | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
| 2 | LMI No. | LMI/OGN/OPS/CHEM/017 Issue No 1 Rev. No.: 04 |
| 3 | Reference Document | COS-ISO-00-OGN/OPS/SYST/065<br>Rev. No.: 0, Mar. 2025 |
| 4 | Superseded Document | Supersedes COS-ISO-00-OGN/OPS/CHEM/017 |
| 5 | Prepared By | Name: R. Balakrishna Patnaik |
| 6 | Checked By | Name: DGM (Chem) |
| 7 | Reviewed By | Name: GM-O&M |
| 8 | Approved By | Name: HOP |

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
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| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

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|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## 1.0 INTRODUCTION

Improper start up, shut down and layup practices usually lead to serious damage in the form of pitting, oxidation and corrosion. Corrosive attack during off-load can also render the plant susceptible to subsequent on-load corrosion or fatigue damage. To preserve the long-term integrity of boiler components, require careful chemical control during start up, shut down and layup condition as well. This guidance note covers the above aspects of the plant including the important factors regarding the choice of the optimum procedures.

## 2.0 SUPERSEDED DOCUMENTS

Cycle Chemistry Guidelines for the Start Up, Shut Down and Layup Practices:
LMI/ OPS/ CHEM/ 017 Rev.4, issued in Feb 2023.

## 3.0 DETAILS OF REVISION

a. Monitoring of Condensate CC/DCC for boiler filling and light-up clearance has been incorporated in drum-type units.
b. A new clause has been added for Hot well-deaerator closed circulation
c. Monitoring of the main steam CC for turbine rolling clearance has been added
d. CBD operation guidelines have been updated
e. Guidelines for installing an automatic pressure-reducing system for SWAS samples in units with low sample flow during cyclic loading have been added.
f. Additional PI tags for subcritical and supercritical units
g. Condensate dumping pipeline dimensions for 110/210/250/500/600/800MW units have been added in Annexure 8.

## 4.0 *OBJECTIVE*

The objective of this guideline is:

a. To maximize unit life and minimize forced outage
b. To minimize impurity ingress, generation and transport
c. To minimize scale formation, corrosion and transport of corrosion material
d. It has been observed that trouble free operation and minimum forced outages are achieved by a combination of control of operating stress and optimum steam water chemistry.

## 5.0 *Scope of the LMI:*

The Scope of LMI is for four number of units with Drum type boilers at NTPC Simhadri for their start up, shutdown and Layup practices during shutdown of units.

## 6.0 *STARTUP PROCEDURE*

The startup procedure for subcritical & supercritical units depends on the layup type and duration. The startup procedure is simpler and faster in short term shutdown as compared to intermediate or long-term shutdown. The following guidelines will play an important role in the early stabilization of the recommended chemistry parameters during unit startup:

1. The CPU and SWAS systems should be available for monitoring and maintaining the recommended chemistry parameters during the startup.
2. The following important guidelines should be followed to ensure the availability of steam water samples in the SWAS during startup and normal operation of the unit:
   a) If the unit has undergone overhauling or modification, the boiler-side sample lines (Feed water, Boiler water, Separator tank, Separator tank outlet, Saturated steam, and Main steam) should be carefully flushed up to the primary rack with pressurized boiler water, immediately following the final hydro test of the boiler.
   b) Before charging any sample into the wet panel, it must be flushed through the blowdown line from the primary rack as well as from the wet panel (one by one) with full flow for at least 1-2 hours to remove any debris from the sample line.
   c) When the unit is coming after overhauling or long-term shutdown, all cation columns should be replaced with fresh, regenerated resin. CPU cation resin should be used for the cation columns (if available). A standby column with regenerated resin should always be available for each SWAS sample.
   d) It is also essential that the core monitoring instruments (refer to 'COS-ISO-00- OD/OPS/CHEM/001' for list of analysers) be in service to provide reliable data as early as possible during the unit startup process.
   e) The SWAS chiller with a standby compressor unit should be available.
3. The Express Lab should be available in the SWAS room for analyzing all chemistry parameters required during the startup and normal operation of the unit.
4. An online total iron analyzer should be used to monitor total iron in the feed water sample during the startup of a supercritical unit.
5. The startup cleanup time depends on the draining rate of open flushing and the circulation rate through the CPU for closed-loop flushing processes.
6. The condensate dumping arrangement, as specified in Annexure-8, should be available for draining high CRUD/TDS water into the atmosphere during startup. If condensate dumping line to atmosphere is not available, it should be provided during OH /Opportunity shutdown.

## 7.0 **Startup procedure for drum type units (sub-critical units)**

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

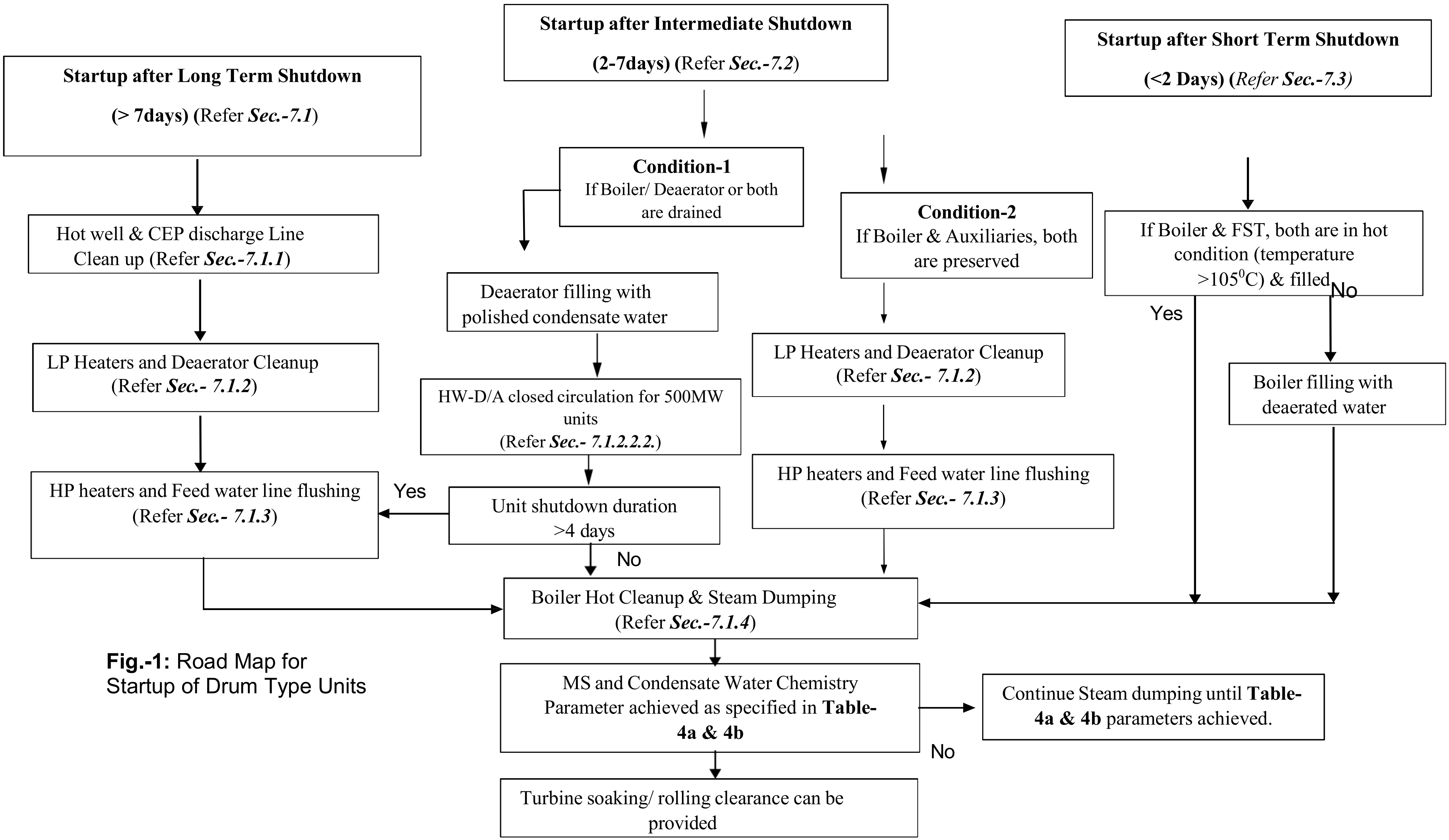


**Fig.-1:** Road Map for Startup of Drum Type Units

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## 7.1 AFTER LONG TERM SHUT DOWN- AFTER OVERHAULING/ MODIFICATION/ RESERVE SHUTDOWN (SHUTDOWN DURATION > 7 DAYS).

The startup of units after a long-term shutdown (Overhauling, boiler modifications, long USD, etc.) involves the following steps:

i) If the boiler is already filled (for wet preservation, hydro testing, etc.), it should be completely drained after thorough mixing with the boiler circulating pump (BCP). If the boiler does not have BCP, it can be drained directly.

ii) All CPU service vessels should be available for operation. If the unit is coming after an overhaul or shut down more than more than 15 days, all CPU service vessels must be charged with fresh resins, and a lot of fresh resin should be available in the regeneration area for changeover.

iii) During all clean-up activities, only one CPU should be kept in service, as each CPU is designed to handle 50% of the TMCR flow in a drum type unit. The bypass valves (both, manual and pneumatic) should be kept fully closed. This guideline is also applicable to start-up after an intermediate or short-term shutdown.

Refer to Annexure: "SOP for Service of CPU Vessel During Unit Startup & Normal Operation" in TC document COS-ISO-00-OGN/OPS/CHEM/20, titled "Guidelines on Operating Philosophy of Condensate Polishing Units."

iv) Refer to OGN-OPS-SYST-056 "Guidelines to Control Steam Purity in Boilers After Overhaul/Modification/Boiler Cold Start-Up" for additional guidelines.

### 7.1.1. HOT WELL and CEP DISCHARGE LINE CLEAN UP

***Objective:*** *To achieve hot well and CEP discharge water quality for charging of CPU (if provided) and flushing of the condensate circuit.*

a) If the hot well is filled with water, start the CEP pump in recirculation. After 15-20 minutes of pump operation, check the quality of the circulating water sample. If it is as per **Table-1**, then condensate dumping to the atmosphere is not required.

b) All three CEP pumps should run one by one during this activity to flush their headers.

c) The CPU should not be put into service during this activity. The deaerator makeup control valve (from the hot well to the deaerator line) should be closed during this process.

d) If hot well water chemistry parameters are not in limit (as specified in **Table-1**), start condensate dumping to atmosphere and keep continuing until condensate dumping water quality achieved as specified in **Table-1**. The condensate dumping line size should be adequate (refer to **Annexure-8** for details) to allow for faster cleanup by dumping the required quantity of water.

e) If the hot well water turbidity is high (>10 NTU) and shows an increasing

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trend, partial or complete replacement of the hot well water with fresh DM water should be considered.

f) A general schematic of the hot well and CEP discharge line cleanup is provided in **Fig. 2**.

**Table-1.** Sample Name: **Condensate Dumping Water**

| **Parameters** | **Limiting Values** | |
|---|---|---|
| | Units without CPU | Units with CPU |
| Silica (ppb), max | 50 | 100 |
| Turbidity (NTU), max | 2 | 5 |
| CC/DCC* (μS/cm) at 25 $^0$C, max | 1 | 1 |
| Chloride*, ppb, max | 20 | 20 |

Note: *: The condensate water cation conductivity (preferably DCC) and chloride levels should be analyzed under vacuum conditions if the unit is coming back after a condenser tube leakage or similar issues.

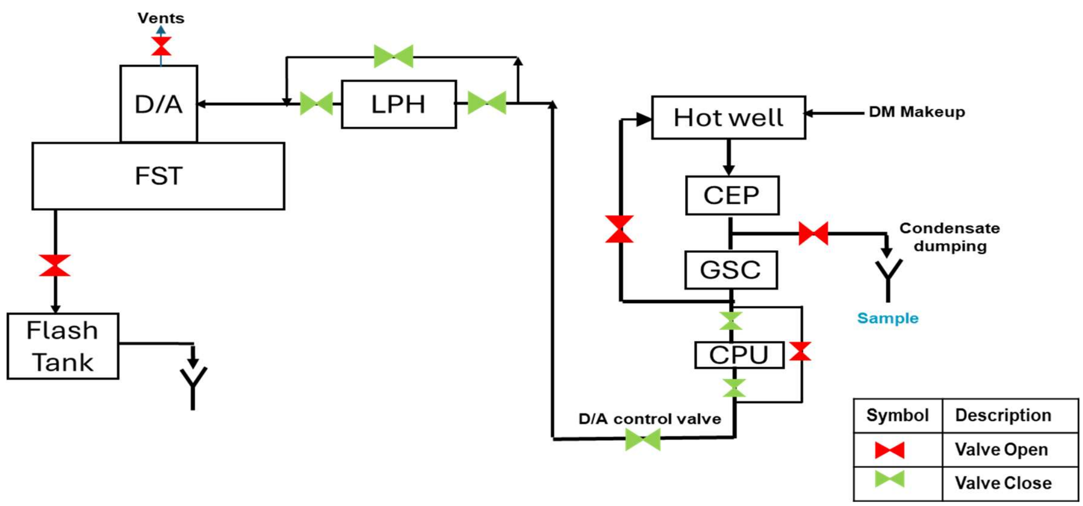


**Fig.-2:** *Schematic for Cleanup of Hot well & CEP discharge Line*

### 7.1.2. LP HEATERS and DEAERATOR CLEAN UP

**Objective***:* To achieve the deaerator water quality for flushing the HP heaters-feed water circuit and subsequently filling the boiler.

i. If the deaerator is filled with water, it should be completely drained. To save time, this can be done simultaneously with the previous step **(Section 7.1.1).**

ii. Ensure that 100% of the condensate water passes through the CPU (preferably through a single CPU to avoid startup loading on all CPUs).

iii. For 500 MW units, both hot well-deaerator open and closed flushing should be carried out.

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### 7.1.2.1. CLEAN UP PROCEDURE FOR 500MW UNITS

#### 7.1.2.1.1. HOT WELL-DEAERATOR OPEN FLUSHING

**a)** Start ammonia dosing at CPU outlet to maintain the FST water pH at 9.3-9.4.

**b)** Start flushing of LP heaters bypass (only *if LP heaters are wet preserved or unit coming after overhaul*) & water side of LP heaters, one by one through deaerator/ feed storage tank & finally drain to atmosphere ***(Fig.-3a).*** Drains of all LP heaters should be opened during this process.

**c)** This clean up step shall be continued until FST drain water chemistry parameters achieved as specified in **Table-2a**.

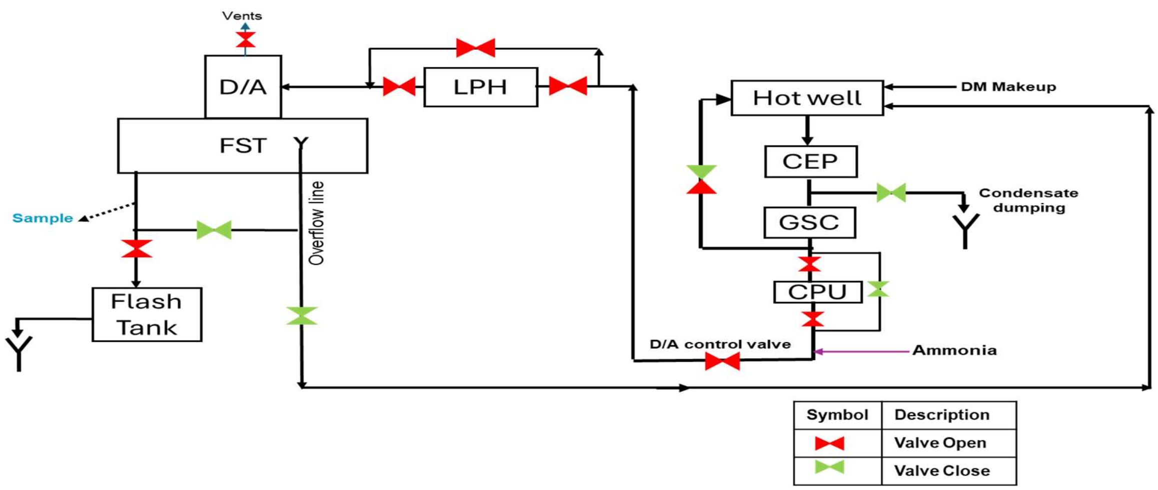


***Fig.-3a**: Schematic for Open Cycle Cleanup of LP Heaters and Deaerator*

#### 7.1.2.1.2. HOT WELL-DEAERATOR CLOSED CIRCULATION

**a)** Stop draining water from the FST to the atmosphere and fill the deaerator to the normal level using condensate water through the CPU. The condenser vacuum should not be pulled during this circulation, and FST heating and pegging should also not be done during this step.

**b)** Start Hot Well - FST - Hot Well closed loop circulation for cleanup as specified in **Fig.-3b**, without draining water to the atmosphere.

**c)** For effective and faster cleanup, the deaerator-to-hot well valve should be kept fully open to increase the circulation rate through the CPU during the closed-loop cleanup.

**d)** This cleanup step should be continued until the FST drain water chemistry parameters (except for DO value) are achieved as in **Tables-2a & 2b**.

**e)** After completing this step, fill the deaerator and start heating the FST to raise the temperature >105°C. Measure the dissolved oxygen in the deaerator water sample and record the actual value in start-up protocol.

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|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

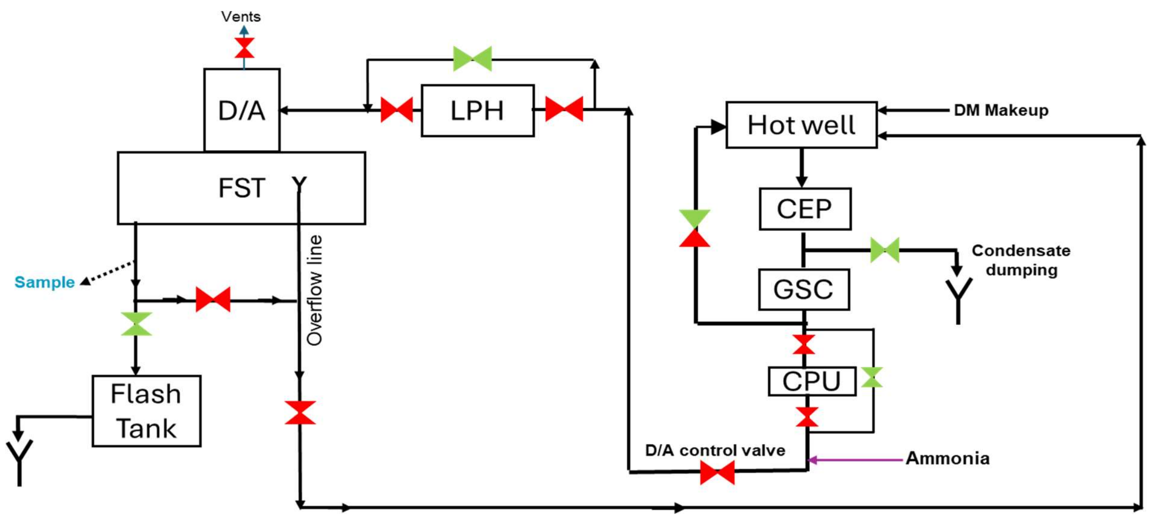


***Fig.-3b**: Schematic for Closed Cycle Cleanup of LP Heaters and Deaerator*

| **Table- 2a** Sample Name: **FST Drain (Deaerator outlet)** | | |
|---|---|---|
| **Parameters** | **Limiting Values** | |
| | **Hot well-Deaerator open flushing** | **Hot well-Deaerator Closed Circulation** |
| pH at 25 $^0$C | < 9.6 | 9.2-9.6 |
| Silica (ppb), max | 100 | 30 |
| Turbidity (NTU), max | 5 | 1.0 |
| CRUD (ppb), max | -- | 50 |
| Total Iron (ppb), max (dissolved & suspended) | -- | 50 |
| DO* (ppb), max | -- | 100 |
| **Note:** The dissolved oxygen (DO) at the D/A outlet should be measured after heating FST >105°C. Refer to **Annexure-10** for the recommended sampling points for offline monitoring of DO. | | |

| **Table-2b:** Sample Name: **CEP discharge** | | |
|---|---|---|
| CC/DCC (μS/cm) at 25 $^0$C, max | -- | 1.0 (downtrend) |
| **Note:** Condensate DCC value should preferably be used for clearance | | |

**Note:**

*i.* Refer to **Annexure-4** for CRUD testing procedure.

*ii.* The sample valve of only the running boiler feed pump (BFP) should be opened to obtain a representative sample of the deaerator in SWAS.

*iii.* CEP discharge, CPU O/L, and deaerator inlet/outlet samples, along with their analyzers, shall be charged in SWAS for monitoring of online chemistry parameters.

*iv.* Where a permanent sample line or point is not available, samples should be taken from the nearest point of the source line to avoid contamination from a long sampling line. The location of the temporary sample point should be safe and easily accessible.

### 7.1.3. HP HEATERS & FEED WATER LINE FLUSHING

**Objective:** To flush HP heaters & feed water circuit through boiler bottom ring headers.

**a)** Ensure that the condenser system is under vacuum and that the condensate CC and DCC values are on a downward trend (refer to **Tables 2a and 2b** for limits). SWAS online analyzers should be used to monitor CEP CC/DCC trends on a continuous basis. The deaerator-to-hot well circuit should be closed before starting the vacuum pulling.

**b)** After achieving FST temperature > 105 $^{0}$C and chemistry parameters as specified in **Tables-2a & 2b**, start flushing of HP heaters and feed water circuit through boiler bottom ring header or LP drain headers (as applicable).

**c)** The bottom ring header drain valves and other drains of the boiler should be fully opened. This flushing activity will ensure the removal of contaminants like silica, loose rust, etc. from the feed water circuit.
Flushing of the feedwater circuit through the bottom ring header should be continued for a minimum of one hour at the maximum possible draining rate. The flushing time can be extended for better results, especially when the units are coming back after an annual overhaul or long USD/FO or boiler modifications.

**d)** DM makeup is to be taken in hot well only, and 100% condensate flow through the CPU is to be ensured.

**e)** Maintaining a good condenser vacuum is essential for controlling dissolved oxygen in the condensate water. Effective oxygen control during startups significantly reduces corrosion product transport.

**Note:** Feed water sample is to be charged in SWAS for monitoring of its online chemistry parameters.

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### 7.1.4. BOILER HOT CLEAN UP & STEAM DUMPING

***Objective**: To achieve Main Steam Chemistry Parameters for Turbine Rolling.*

a) Ensure that the condenser system is under vacuum. Additionally, a downward trend in condensate CC/DCC (refer to **Tables 2b & 3b** for limits) is required for boiler light-up clearance. Boiler light-up should not proceed if condenser tube leakage or seepage is suspected.

b) Boiler hot cleanup begins after the first BLU.

c) If unit is coming back after overhaul or boiler modification, boiler must be completely hot drained at 10-15 kg/cm² drum pressure after the first BLU.

d) The boiler should be refilled with hot deaerated water (temperature >105°C and chemistry parameters as in **Tables-2a & 2b/** for next light-up.

e) Maintain the boiler water pH and conductivity within the recommended operating range by using LP dosing and carefully dosing Sodium hydroxide to the boiler drum at approximately 0.10-0.50 ppm. **Excessive Caustic dosing should be avoided.** The auto-stop logic for the HP dosing pumps should be in place in all units.

f) The boiler drum blowdown should be maintained throughout the startup period until the contaminant levels are within the specified limits.

g) The condensate dumping valve should preferably be fully opened before charging the HP-LP bypass line.

h) For units with CPU coming after overhauling/ boiler modifications: After charging the HP-LP bypass line, if the turbidity of the condensate water abruptly increases (>5NTU, with an increasing trend), DM makeup can be taken into the FST from the boiler fill pump. The deaerator control valve should be closed and ensure that no condensate flow between the hot well and FST. However, the CPU should remain in service with the bypass valve fully closed. Keep the condensate dumping valve fully open to dump the water to atmosphere and monitor the dumping water quality (Silica and Turbidity) at 30-minute intervals. The recommended FST temperature is to be maintained always for removal of dissolved oxygen.

   Once the condensate water chemistry parameters are within the desired range, with silica <100 ppb and turbidity <5 NTU, DM makeup should be restored to the hot well, and the condensate dumping rate should be adjusted based on these chemistry parameters. The availability of boiler fill pump line to the deaerator should be ensured during the unit overhaul.

i) Ensure that 100% of the condensate water passes through the CPU (preferably through a single CPU to avoid startup loading on all CPUs). A second CPU can also be taken into service if the first cannot handle the total condensate flow.

j) Monitor and record the individual CPU flow, differential pressure of the vessel and resin traps, and online chemistry parameters in the CPU logbook. PI monitoring tools be used for monitoring of CPU parameters.

k) Condensate water, CPU outlet water, feed water, boiler water, and steam water samples should be analyzed once/two hour, for pH, specific conductivity, turbidity, silica, and sodium (boiler).

l) Deaerator outlet and feed water silica trend also to be kept under close observation and maintain <30ppb throughout the startup process.

m) Boiler water, saturated steam, and main steam samples should be charged to SWAS for monitoring their online chemistry parameters.

n) The boiler hot cleanup and steam dumping schematic is given in **Fig.-4**.

o) Once main steam & condensate parameters (as specified in **Tables-3a** and **3b**) are achieved (temperature & pressure) & show continuous downtrend, turbine rolling clearance shall be accorded by Chemistry.

p) The drum level is to be carefully controlled to minimize the potential for sodium hydroxide to be carried over into the steam.

q) If in-service CPU is used since the initial cleanup, it should be replaced with a fresh resin lot before or immediately after the start of turbine rolling. During the resin transfer period, the second CPU must be in service, and the bypass valve should be kept fully closed if the total condensate flow is less than 50% of TMCR flow. The differential pressures of the CPU service vessels and resin traps are to be monitored and recorded.

r) When a unit is coming back after an overhaul, boiler modification, or long USD, the HP heaters drip line should be flushed to the condenser before being directed to the deaerator.

s) Reduce or stop the boiler blowdown when the boiler and steam chemistry parameters are within the normal limits as specified in COS-ISO-00-OD/OPS/CHEM/001.

t) When condensate dumping water quality is in recommended limit of CPU (refer **Table-1**), minimize or stop condensate dumping.

u) During startup, any excursions in chemistry parameters must be addressed quickly & effectively, and a root cause analysis should be conducted for remedial action & observations should be recorded in the unit startup protocol **(Annexure-1A**).

v) A startup checklist/ protocols (**Annexure-1A** & **Annexure-1B**) to be jointly signed by representatives of Chemistry and Operation. **Annexure-1B** should be signed by Chemistry only. The protocols are to be uploaded and maintained on the site intranet by Chemistry.

w) Ensure 100% condensate flow through the CPU during startup and the normalization of unit chemistry parameters. The second CPU can be taken out of service (not applicable to seawater-cooled units) **48 hours** after unit synchronization.

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**Note:**

*i.* If the unit has returned after an overhaul, caustic dosing in the boiler should be done only after the completion of the boiler safety valve floating activities. For mixed metallurgy units, TSP dosing in the boiler drum should be done cautiously to maintain the boiler water pH between 9.2-9.6. The phosphate level in the boiler should be checked and recorded.

*ii.* During the safety valve floating, ensure careful monitoring of the chemistry parameters for both the MS and condensate samples.

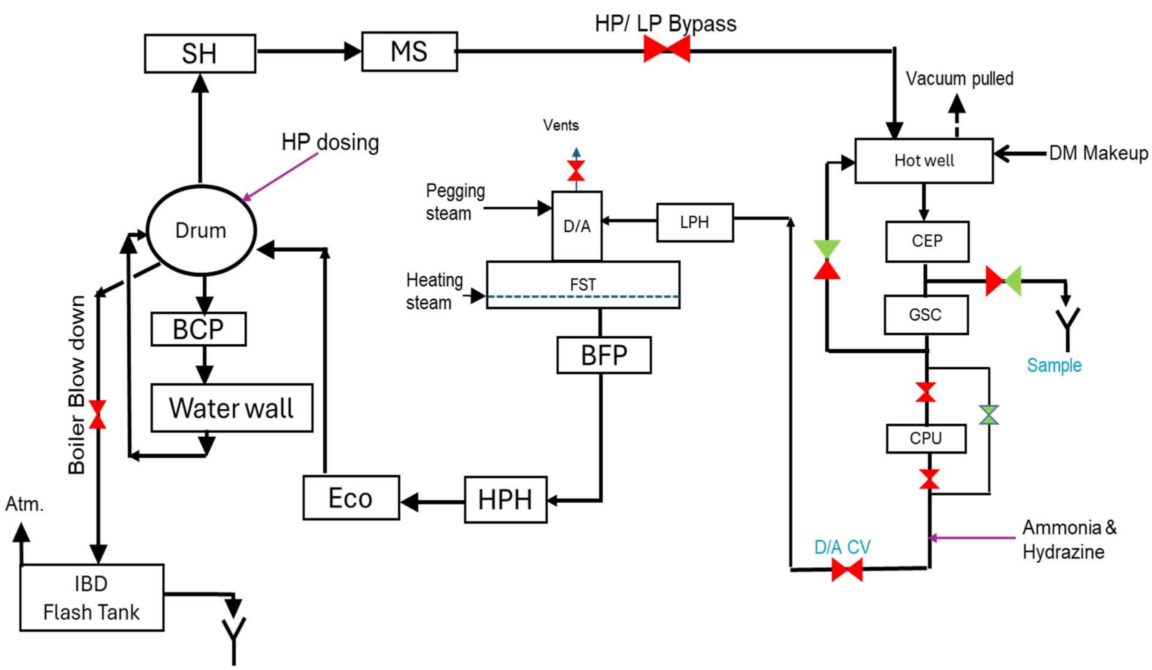


**Fig.- 4:** *Schematic for* Boiler Hot Cleanup and Steam Dumping

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**Table-3a** Sample Name: **Main Steam**

| **Parameters** | **Limiting values** | |
|---|---|---|
| | For mixed metallurgy unit | For all ferrous unit |
| pH at 25 $^0$C | 8.8-9.3 | 9.2-9.6 |
| Silica (ppb), max | 20* | 20* |
| Sodium (ppb), max | 5 | 5 |
| CC (µS/cm) at 25 $^0$C, decreasing trend | --- | 1.0# |
| Total Iron (ppb), max (dissolved & suspended) | 10 | 10 |
| CRUD (ppb), max | ---- | Nil** |

**Table-3b** Sample Name: **Condensate water**

| **Parameters** | **Limiting values** | |
|---|---|---|
| | For mixed metallurgy system | For all ferrous system |
| CC/DCC (µS/cm) at 25 $^0$C | <1.0 (downtrend) | |

**Note:** Since condensate water turbidity is expected to be high during the initial steam dumping period, the condensate CC analyzer should remain charged, and the DCC analyzer can be charged after a decrease in the condensate water turbidity and before the start of turbine rolling.

**Note:** *: For turbine rolling clearance, the main steam silica level should be <20 ppb. However, during the ramp-up of the unit, silica in the main steam can be maintained up to 30 ppb for a maximum of 12 hours after synchronization of the unit. This should be recorded in the startup protocol. However, the station should always maintain 100% condensate flow through the CPU and reduce the main steam silica to <20 ppb at the earliest by performing sufficient blowdown from the boiler, along with frequent sampling and analysis of steam water sample (at 30-minute intervals)

#: For 500 MW units only **: The CRUD paper should be colorless

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## 7.2 AFTER INETRMEDIATE SHUT DOWN (SHUTDOWN DURATION 2-7 DAYS)

### 7.2.1. CONDITION-1

If the boiler, deaerator, or both are cold or drained for maintenance, the unit startup sequence shall proceed as follows:

**Fill the deaerator with condensate water**

*(condenser under vacuum and 100% condensate flow through CPU)*

↓

**Hot well-Deaerator closed circulation for 500MW units only**

*(Section-7.1.2.1.2)*

↓

FST heating >105 $^{0}$C

*(Section- 7.1.2.1-d/ 7.1.2.2.2-e)*

↓

HP heaters and feed water line flushing

*(If startup period >4days, Section- 7.1.3)*

↓

Boiler hot cleanup & steam dumping

(*Section- 7.1.4*)

### 7.2.2. CONDITION-2:

If the boiler and auxiliaries are preserved, follow the steps specified in *Section-7.1.2, 7.1.3* & *7.1.4*

## 7.3. AFTER SHORT TERM SHUT DOWN:

(SHUTDOWN DURATION <2 DAYS)

No cleanup cycle is required. However, if the boiler is drained for maintenance purpose, it should be refilled with deaerated feed water (temperature >105°C). Boiler light-up & subsequent activities should be carried out as **Section 7.1.4**.

## 7.4. RECOMMENDATION FOR CONTINUOUS BLOWDOWN(CBD) OPERATION IN DRUM-TYPE BOILERS:

In drum-type units, boiler continuous blowdown (CBD) should be opened at periodic intervals or as needed to maintain the boiler water chemistry parameters within the recommended limits as specified in COS-ISO-00-OD/OPS/CHEM/001.

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The following recommended boiler CBD schedule should be maintained during normal operation:

**a)** In 500 MW units, where the boiler water sample line is taken from the CBD line, the CBD should be open for at least 2 hours every week.

**b)** In 500 MW units, where the boiler sample line is taken from the down comer header, the CBD should be open for at least 2 hours every 15 days.

**c)** In units equipped with a CPU, the CBD should be opened for at least 2 hours immediately after the changeover of the exhausted CPU.

**d)** In addition to the above conditions, the CBD can also be opened, as needed, to maintain the boiler water chemistry parameters within the recommended limits as specified in COS-ISO-00-OD/OPS/CHEM/001.

**e)** CBD data should be recorded on a 100% basis for each unit, along with the reasons.

## 8.0. LAYUP PRACTICES

The shut down and layup period should be regarded as a continuum of the good practices used during operation. The key objective of cycle chemistry is to provide protective oxide surfaces on all components throughout the steam and water circuits whereas the primary purpose of layup practice is to maintain those protective surfaces during offload condition.

During shutdown, a moist, oxygenated environment comes into contact with the deposits and evaporated liquid films that form during operation, resulting in an aggressive local environment.

The layup procedure is generally divided into two categories, i.e., the dry and wet layup.

## 8.1 DRY LAYUP

It requires rapid draining of water-touched components and evacuating the reheat steam lines (using the vacuum in condenser) while the piping is still hot followed by continuous purging with instrument air (dry air) or nitrogen gas to maintain a low moisture environment throughout the system. Following are some minimum requirements for dry air preservation of any equipment with instrument air:

**a)** Equipment should be completely drainable.

**b)** Instrument air (dew point < -15 $^0$C at atmospheric pressure) at filling points.

**c)** Air supply line connection & sampling points as per approved drawing.

**d)** Ensure the availability of pressure gauge(s) or transmitter(s) at filling points to monitor the air pressure of the preserved equipment.

### 8.2 WET LAYUP

It requires filling most of the system with an alkaline solution containing ammonia and hydrazine (hydrazine for mixed metallurgy units only) and preventing air ingress by nitrogen capping. Hydrazine (reducing chemical) should not be used for boiler preservation where feed water chemistry is AVT (O)/OT (all ferrous units).

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## 9.0 LAYUP PROCEDURE FOR BOILER & AUXILIARIES:

The equipment layup matrix for all types of boilers & auxiliaries is given

**Table-4** Equipment Layup Matrix for Drum-type and Once-through Boilers & Auxiliaries

| Shut-down duration | SYSTEMS | | | | |
|---|---|---|---|---|---|
| | Hot well | Deaerator | LP & HP heaters | Drum / Supercritical Boilers Including Super heaters | Reheaters (RH) |
| **08-12hrs (Hot standby)** | • Maintain Vacuum and turbine seals. | • Keep deaerator temp >110 $^0$C and pressurized PRDS/ Auxiliary steam to maintain DO <100 ppb in FST water.<br>• Maintain the same chemistry parameters as that of normal operation. | • Leave it wet (tube side), try to maintain DO <100 ppb.<br>• Maintain the same Chemistry parameters as that of normal operation. | • Keep boiler completely box up till pressure in Steam circuit drops to venting pressure 10 Kg/cm$^2$ and boiler water wall (WW) filled till WW minimum temperature drops to draining temperature approx. 110 $^0$C (Stop fans when FG inlet at APH inlet comes approx. 220 $^0$C.)<br>• Maintain the same Chemistry parameters as that of normal operation.<br>• | • Please refer to **Section 9.1.1.1** for dry air preservation of reheater circuit. |
| **12 hrs-4 days** | • Maintain Vacuum and turbine seals.<br>• If there is no vacuum, keep the hot- well as | • Keep deaerator temp >110 $^0$C and pressurized by PRDS/ Auxiliary steam (Refer ***Section- 9.2.3.1***)<br>• If PRDS/ auxiliary steam is not available, then maintain the same chemistry parameters as | • **Shell side**: Open vents and drains and execute dry layup (Refer ***Section- 9.1.2.2/9.3***)<br>• **Tube side:** Leave it wet and maintain the same chemical parameters as that of normal | • As specified above.<br>• **DRY WITH INSTRUMENT AIR:** When main steam pressure reaches 10 Kg/cm$^2$, open SH vents and subsequently drum/ separator vents at 2 Kg/cm$^2$. When WW temp. reaches approx. 110$^0$C drain complete boiler. Start instrument air purging when platen coil achromat temp comes 110 $^0$C or if achromat is not available, SAPH flue gas I/L temperature reaches 110 $^0$C or steam stops coming from vent. (Refer ***Section- 9.1.1.2***). | • Keep maintaining purging air relative humidity <35% throughout the preservation period. |

| | | | | | |
|---|---|---|---|---|---|
| | such (pH>9.0). | that of normal operation. | operation. **OR** Open vents and drains and execute dry layup (Refer ***Section- 9.1.2.1/9.3***) | • **DRY WITH N2 GAS:** Drain hot under N2 blanket (Refer ***Section 9.3***)<br>• **WET LAYUP:** Maintain the same chemical parameters as that of normal operation. The nitrogen should be applied while the superheater/drum is still under positive pressure. | |
| **>4 days** | Drain hot well & keep dry. (Refer **Sec.-9.1.4**) | • **WET LAYUP** (Refer **Section- 9.2.3.2**) | • **Shell side**: Continue dry layup<br>• **Tube side:** Continue dry layup or execute wet layup (Refer ***Section- 9.1.2.1***). | • **DRY LAYUP:** Continue dry layup. **or**<br>• **WET LAYUP:** Backfill the super heater and raise the boiler pH (as specified in **Table-6**) and keep capped by N2/ pressurized to prevent air ingress in the system. (Refer **Section-1.2.1.2**). | |

**Note:**

- **Dry air lay- up is always a good choice for faster startup of unit.** Instrument air at ambient temperature can be filled for dry preservation.
- If maintenance access is required to boiler, deaerator or other steam cycle equipment that normally contains water/steam, dry layup will be good option.
- Hydrazine should not be used for preservation of boiler & auxiliaries where feed water chemistry is AVT (O)/ OT.
- **Reheater to be always kept under dry preservation. If reheater bends are dry, then preserve with instrument air and if not sure then by N2 gas.**
- All layup conditions, wet or dry, should be either continuously or periodically monitored.

### 9.1 DRY AIR (INSTRUMENT) AIR) LAYUP PROCEDURE

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|---|---|
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a) Dry layup with instrument air is the most preferred preservation scheme for faster startup. The process of drying the boiler tube and pipe internals by **latent heat is very important in dry air preservation,** so the boiler should be drained when the minimum water wall temperature reaches approx. $110^0$C.

b) The dew point of purge-in instrument air should be < -15 $^0$C at atmospheric pressure at filling point(s) (a well-maintained instrument air system should produce a dew point of air at about -40 $^0$C at atmospheric pressure), and the relative humidity (RH) of purge-out air should be about < 35%. The relative humidity should be monitored at several locations (lowest drain points) to determine the status of the dry air preservation.

c) Monitoring of dew point of purge-in air and relative humidity of purge-out air shall be done Once/day or more depending on weather conditions.

d) During dry air preservation of any equipment, inside air pressure should be maintained in a range of 0.3-0.5 $kg/cm^2$ (just above the atmospheric pressure) as relative humidity is directly related to air pressure.

e) Dry layup with instrument air not to be done at Sea shore station.

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### 9.1.1 PRESERVATION OF BOILER, SUPER HEATER & RE-HEATER CIRCUIT

#### 9.1.1.1. REHEATER PRESERVATION

a. Reheater dry air preservation is to be started just after shutdown of the unit. For depressurization of the reheater, follow guidelines as for superheaters.

b. Allow instrument air to flow through the reheater to the condenser at least 30 minutes before killing the condenser vacuum.

c. After killing the vacuum, keep continuing instrument air flow through reheater circuits as specified in schematics Fig. 20 & 22

d. After injecting instrument air, measure the relative humidity at the lowest drain points of each circuit and maintain RH <35% to avoid any dew formation in preserved equipment/circuits.

e. Pressurize the boiler internal and other circuits with instrument air and maintain pressure in a range of 0.3-0.5 kg/cm2 (just above the atmospheric pressure).

f. Measure and record the dew point of purge-in air, relative humidity of purge-out air, and air pressure of the system daily.

g. Only dry preservation is recommended for the reheater circuit. If reheater bottom bends are not dry due to condensation of steam, then N2 gas (refer to Sec. 9.3) preservation or dry preservation followed by boiler light-up (refer to Sec. 9.4) is to be done.

#### 9.1.1.2. BOILER AND SUPERHEATER PRESERVATION:

a. The main steam temperature decreased at a rate of 50-60 $^0$C per hour after boiler shutdown.

b. When main steam pressure reached 10 Kg/cm2 in pendant type SH (non-drainable SH) and 5 Kg/cm2 in horizontal type SH, open all vents valves and eject the steam in tubes and pipes. Drum/separator vents to be opened at 2 Kg/cm2.

c. When the water wall temperature becomes 110 0C, open the water wall section drain valves (the latent heat of WW will be used to dry WW) and then drain all the water from the boiler WW section.

d. If thermocouple is not there it should be mounted on water wall firing floor near corner 1.

e. Complete box-up of the boiler is to be ensured after stoppage of fans (APH FG inlet temperature < 220 0C) and seal trough sealing to be maintained. All manholes and peepholes are to be kept closed to ensure the drying of the boiler heating surfaces by the latent heat of the component itself

f. After completion of draining, start instrument air purging when minimum achromats temperature reaches approx. 110 0C or if achromat is not available, SAPH flue gas I/L temperature reaches 110 0C or steam stops coming out from vent line as per respective circuit schemes (Refer to Annexure-3 & Annexure-4). Instrument air at ambient temperature can be filled for dry preservation. Arrangements are to be made for the instrument air pipeline to be routed through the penthouse to supply hot air. This is preferable but not necessary to start with.

g. Pressurize the boiler's internal and other circuits with instrument air and maintain pressure in a range of 0.3-0.5 kg/cm² (just above the atmospheric pressure). Maintain a partial opening of header drains and vents at the opposite end (if drain on LHS is opened then vent of RHS is to be used for filling to avoid air short circulation), at least one drain and one filling line in all circuits, to establish flow of instrument air and to control the boiler internal air pressure.

h. After injecting instrument air, measure the relative humidity at the lowest drain points of each circuit and maintain RH <35% to avoid any dew formation in preserved equipment/circuits.

i. Measure and record the dew point of purge-in air, relative humidity of purge-out air, and air pressure of the system daily:

   I. Furnace front wall inlet header drain line/ low point drain for both drum type and once- through boilers.
   II. Economizer inlet drain or any other suitable location.
   III. Hot reheat line drain valve left & right side.
   IV. Any other suitable location based on local conditions.

j. The innermost loop of platen SH/final SH is to be checked for any condensation by radiography/coil heating, and in case of any water in the loop, instead of dry, only wet preservation is to be employed.

   I. Schematic drawing for instrument air preservation of drum type boiler & Super heater circuit (Annexure-2, Fig.-9).
   II. Schematic drawing for instrument air preservation of reheater circuit of drum type boiler, (Annexure-2, Fig.-10).
   III. The above schematics are general guidelines for instrument air preservation of boilers, superheaters, & reheaters. A station-specific schematic drawing, based on the above guidelines & local conditions, shall be prepared by the site and may be vetted by the COS-Boiler group before its implementation.

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9.1.1.3. **SCHEMATICS**

Note:

***i.*** For deciding instrument air dry preservation of water wall and SH circuit (after hot draining of boiler, refer Section-9.1.1.2c), Platen SH achromats minimum temperature should be more than 110 0C (when achromat temperature is available) or flue gas I/L temperature at SAPH I/L to be > 110 0C or steam coming out of SH vents otherwise boiler wet preservation is to be done.

*ii.* If Boiler platen SH minimum achromat temperature has become less than 110 $^0$C (when achromats temperature is available) or SAPH flue gas inlet temperature is < 110 $^0$C and steam not coming from SH vents then for instrument air preservation of boiler, first it is to be reheated by boiler light-up to achieve 220 0C SAPH flue gas I/L temperature and then hot draining of boiler followed by filing of instrument air (refer Section-9.1.1.2).

*iii.* For already cooled boiler dry preservation should not be attempted due to condensed water in pedant coils and low-level water pooling points.

9.1.2. **PRESERVATION OF HEATERS**

9.1.2.1. **WATER SIDE (TUBE SIDE) OF HEATERS (Preferably for HP Heaters)**

a) If heater tube metallurgy is stainless steel and heater is completely drainable (both conditions must be ensured at site before going for instrument air preservation of heaters), then dry air preservation of water side (tube side) can be done with instrument air. The feed water trains on tube side shall be isolated by closing inlet and outlet valves.

b) Open vents and drains of all heaters in hot condition to ensure complete drying of the heaters' water side surfaces by the latent heat of components itself. OEM recommendations should be followed for hot draining of the tube side of heaters.

c) Start continuous purging of instrument air from the vent of each heater to achieve relative humidity <35% at their low point drains. Instrument air dew point at filling point should be < -15 $^0$C at atmospheric pressure.

d) Pressurize the tube side of heaters with instrument air and maintain pressure in a range of 0.3-0.5 kg/cm2 (just above the atmospheric pressure). Maintain partial opening of each drain to establish flow of instrument air and to control internal air pressure.

9.1.2.2. **SHELL SIDE OF HEATERS**

a) Shell side of heaters should be preserved only by instrument air or $N_2$ gas. Open vents and drains of all heaters in hot condition to dry the internals. Shell side condensate & steam can also be evacuated by condenser vacuum.
b) Start continuous purging of instrument air from vent to achieve relative humidity < 35% at low point drain of each heater. Instrument air dew point at filling point should be < -15 $^0$C at atmospheric pressure.
c) Pressurize the shell side of heaters with instrument air and maintain pressure in a range of 0.3-0.5 kg/cm$^2$ (just above the atmospheric pressure). Maintain partial opening of each drain to establish flow of instrument air and to control internal air pressure. A dry preservation facility should be available.

9.1.3. **PRESERVATION OF DEAERATOR**

Generally, deaerator is preserved in wet condition as trial of BFP is to be done on regular intervals.

9.1.4. **PRESERVATION OF HOT WELL**

Completely drain the hot well and keep it dry. During turbine preservation, measure the relative humidity of the hot well drain air to ensure dryness.

## 9.2 WET LAYUP PROCEDURE

Wet layup involves maintaining water in the boiler, deaerator, and hot well while excluding oxygen from the equipment to prevent corrosion. It is often the method of choice when a unit needs to be returned to service on short notice or when dry layup facilities are not available.

The filling water quality for wet layup is specified in ***Table-5***. Hydrazine (for mixed metallurgy units) and ammonia should be added in a manner that results in a uniform concentration throughout the system. After filling the system, establish and maintain 0.3-0.5 kg/cm$^2$ nitrogen cap on the vents of super heater, steam drum/separator and other wet preserved system.

Stagnant conditions should be regarded as a prerequisite condition for the inception of pitting in carbon steels at wetted surfaces when oxygen is present. Therefore, to avoid pitting and other related localized corrosion on metal surface, wet preserved equipment should be circulated on a regular interval as prescribed in this document and top-up of chemicals to be done, if required.

| **Table-5 Feed** (Filling) Water Quality for Wet Layup |
|---|

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| **LMI No.** | **LMI/OGN/OPS/SYST/065, REV:0** |

| SHUTDOWN DURATION | ALL FERROUS SYSTEM<br>FW regime: AVT-O/OT | MIXED METALLURGY SYSTEM<br>FW regime: AVT(R) | |
|---|---|---|---|
| < 4 days | Maintain the same chemistry parameters as that of normal operation. | | |
| > 4 days | pH: 10.0-10.5 | **For equipment containing Copper alloy**<br>(Condenser, feed water heaters, etc.) | **For equipment containing only ferrous metallurgy**<br>(Boiler, Superheater, etc.) |
| | | pH: 9.0-9.2<br>Hydrazine: 40-50 ppm | pH: 9.5-10<br>Hydrazine: 200-250 ppm |
| | Filling water turbidity should be <1 NTU | | |

#### 9.2.1 PRESERVATION OF BOILER & SUPERHEATER CIRCUIT

For optimum storage of the boiler, increase blowdown on shutdown to achieve minimum concentrations of contaminants (like phosphate, chloride, silica, etc.) in the boiler water of a drum-type unit.

##### 9.2.1.1 SHUTDOWN DURATION < 4 DAYS

**a)** After the unit shutdown, keep the boiler above the saturation temperature and pressurized for as long as possible.

**b)** The nitrogen should be applied while the superheater/drum is still at positive pressure (for example, ~2 kg/cm$^2$). As the steam pressure decays, nitrogen fills the steam space. Maintain 0.3-0.5 kg/cm$^2$ nitrogen cap.

##### 9.2.1.2 SHUTDOWN DURATION > 4 DAYS

**a)** For Once-though boiler, super heater can be filled by overflowing of separator (separator water turbidity <1NTU, to be checked after running of BCP) after complete mixing of ammonia solution in boiler. Filling water quality is specified in ***Table-5***. Boiler & Super heater vents samples pH should be 10-10.5.

**b)** For drum type units, super heater circuits should be filled only by backfilling line and not be filled by flooding the boiler drum into the superheater.

If the boiler uses any type of solid alkali treatment (phosphate or caustic) during normal operation, overflowing the boiler to fill the superheater will contaminate both the superheater and turbine during subsequent operation.

**c)** If the superheater backflushing line is not available, the boiler should be filled to the normal drum level and circulated using the boiler circulation pump (BCP, if envisaged in the design). The water should be checked for phosphate, chloride, and turbidity. If phosphate and chloride are below detection limits (BDL) and turbidity is less than 2 NTU, the superheater can be filled by overflowing the boiler water. Otherwise, drain the boiler and repeat the flushing process with fresh ammoniated DM water.

**d)** After completion of super hater filling, start boiler filling with feed water of the quality specified in ***Table-5*.**

**e)** Care must be taken not to exceed permissible DT between boiler metal temperature/ Super heater metal temperature and feed water temperature as prescribed by OEM recommendations.

**f)** Establish and maintain 0.3-0.5 kg/cm$^2$ nitrogen cap at the vents of super heater, steam drum / separator and other preserved system after completion of filling.

**g)** If $N_2$ capping facility is not available, then keep boiler always pressurized up to 10 kg/cm$^2$. Feed water quality (being used for pressurization) shall be as per ***Table-5***.

**h)** During wet preservation, chemical concentration near surface can be lowered and to avoid pitting corrosion, it is recommended to circulate the boiler **once a week** by boiler circulating pump (if envisaged in design). If chemical concentration is found lower than specified values (refer to ***Table-5***), the chemical mixed water to be injected into the preserved water to restore the desired chemistry parameters. After taking any fresh makeup in the boiler, it should be thoroughly circulated & mixed by boiler circulation pump (BCP) and chemistry parameters to be analyzed and recorded in preservation protocol.

**i)** For units where a boiler circulation pump is not included in the design, boiler water circulation can be maintained using a suitably sized external pump arrangement. An alternative method is to drain a reasonable amount of water from the boiler and refill it with fresh treated water, as specified in **Table-5**, at regular intervals **(preferably fortnightly).** However, circulating the boiler water with an external pump is the preferred option. Boiler water chemistry parameters and the quantity of makeup water should be recorded in the preservation protocol.

**j)** Only equipment requiring maintenance needs to be drained and nitrogen (if used for capping) purged with air to provide an environment suitable for safe entry of maintenance personnel.

**k)** If the boiler is drained for any reason, then boiler shall be refilled with treated water of quality as specified in ***Table-5*** and capped with nitrogen or kept pressurized.

**l)** A detailed protocol is to be filled up as per ***Annexure-3***.

### 9.2.2 PRESERVATION OF WATER SIDE (TUBE SIDE) OF HEATERS

#### 9.2.2.1 SHUTDOWN DURATION < 4 DAYS

Leave water side of heaters in wet and pressurized condition and maintain the same chemistry parameters as of normal operation.

#### 9.2.2.2 SHUTDOWN DURATION > 4 DAYS

The metallurgy of these devices is of paramount importance when determining feedwater chemistry for intermediate and long-term storage. For systems containing copper alloys, hydrazine and ammonia solutions (40-50 ppm of hydrazine and only the required amount of ammonia to achieve a pH of 9.0-9.2) should be used, and oxygen ingress must be avoided to minimize copper corrosion. High concentrations of ammonia should not come into contact with copper alloys. It is essential to maintain reducing conditions (ORP should be between -150 and -350 mV) throughout the wet preservation period.

For all ferrous metallurgy systems, a wet layup is performed with a pH of 10-10.5, achieved with ammonia, and a nitrogen cap of 0.3-0.5 kg/cm² is maintained at the vents of the tube side of heaters. A detailed protocol should be completed as per **Annexure-3.**

#### 9.2.2.3. CIRCULATION SCHEME FOR CONDESNATE & FEED WATER CIRCUIT DURING WET LAYUP

If the unit is to be kept under shut down for a long period and wet layup is the only choice for preservation, then flushing (partial/full replacement of water)/circulation of preservative water should be done minimum ***once in a month*** as per scheme given in ***Fig.- 5-7*** to *avoid pitting and other related localized corrosion on metal surface due to stagnation of water*.

**The LP turbine dehumidifier must be in continuous service to prevent moisture accumulation in the LP turbine from the hot well during circulation.**

#### 9.2.2.3.1. CIRCULATION SCHEME FOR LP HEATERS & DEAERATOR

Take fresh makeup in hot well (if drained) & re-recirculate by CEP (minimum recirculation) & dump to atmosphere until dumping water turbidity reaches <2 NTU.

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Stop condensate dumping & start LP heaters preservative water circulation as per scheme given in ***Fig.-5-a or Fig.-5-b*** depends on deaerator design at site. Start dosing of concentrated solution of hydrazine (if applicable) & ammonia at CEP discharge (chemical dosing provision should be provided if not available). As circulating water ammonia concentration is very high, CPU should not be taken in service during this circulation.

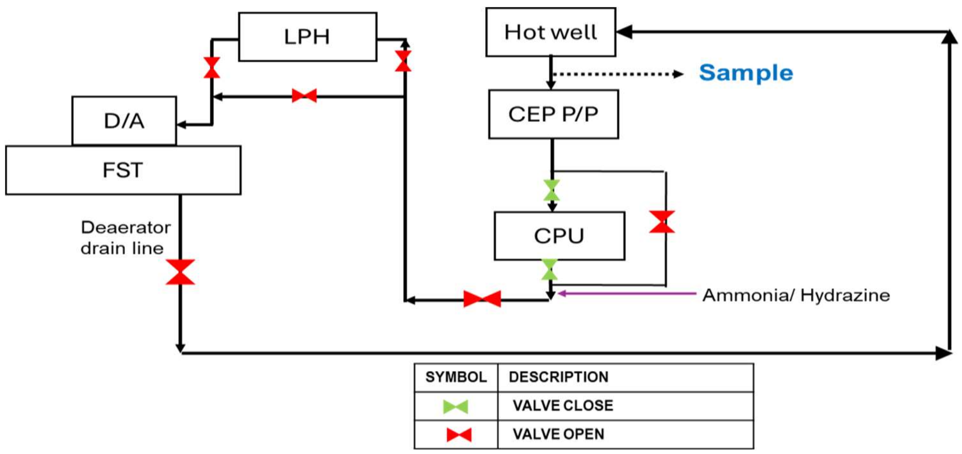


**Fig.- 5-a:** *Schematic for Circulation of LP heaters and Deaerator Water*

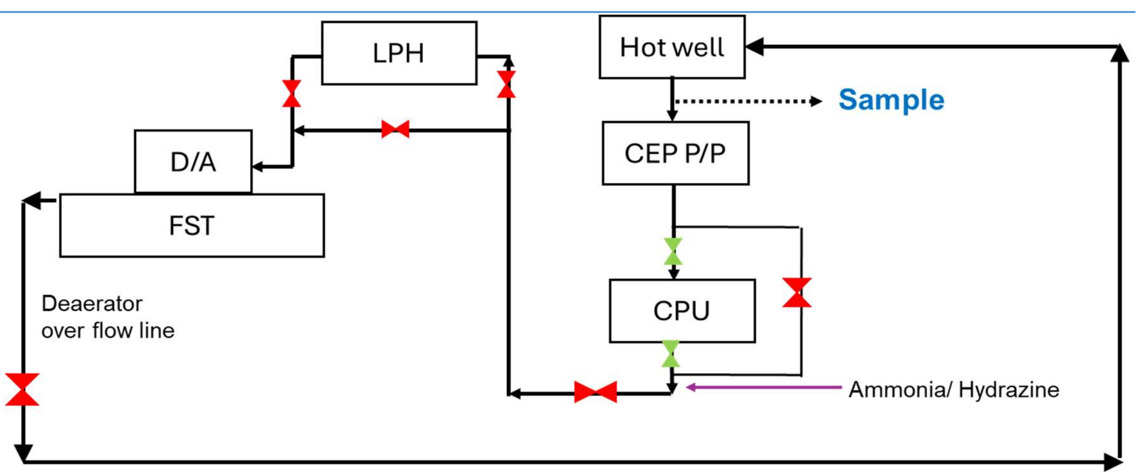


**Fig.- 5-b:** *Schematic for Circulation of LP heaters and Deaerator Water*

If circulation is done according to the above schemes (**Fig.-5-a or 5-b**), then completion criteria should be as per ***Table-6*.**

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|---|---|
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| **Table-6** Samples Name: **CEP Discharge or FST Drain/ Overflow Line** | | |
|---|---|---|
| **Parameters** | **Limiting Values** | |
| | For mixed metallurgy unit | For all ferrous unit |
| pH at 25 $^0$C | 9.0-9.2 | 10.0-10.5 |
| Hydrazine (ppm) | 40-50 | -- |
| Turbidity (NTU), max | 3 | 3 |

If turbidity of circulating water exceeds 3 NTU, it can be controlled either by partial draining and taking freshly make up or by circulation through CPU (if provided).For units, where circulation is not possible, LP heater circuit water can be partially replaced with fresh treated water as per scheme given in ***Fig.-6***. During this activity, CPU may be taken into service (if required). The condensate flow should be minimum to achieve above recommended pH in condensate water. The completion criterion is given in ***Table-7***.

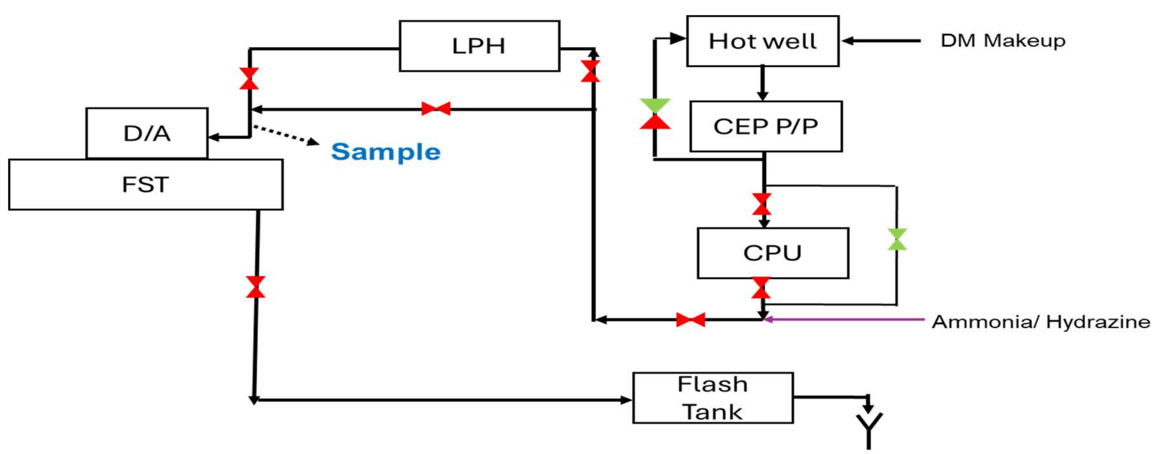


**Fig. 6:** *Schematic for Flushing (replacement) of LP Heaters Water*

| **Table-7** Samples Name: **Deaerator Inlet/ Last LP Heater Outlet** | | |
|---|---|---|
| **Parameters** | **Limiting Values** | |
| | For mixed metallurgy | For all ferrous unit |
| pH at 25 $^0$C | 9.0-9.2 | 10.0-10.5 |
| Hydrazine (ppm) | 40-50 | -- |
| Turbidity (NTU), max | 3 | 3 |

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
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#### 9.2.2.3.2 FLUSHING SCHEME FOR HP HEATERS

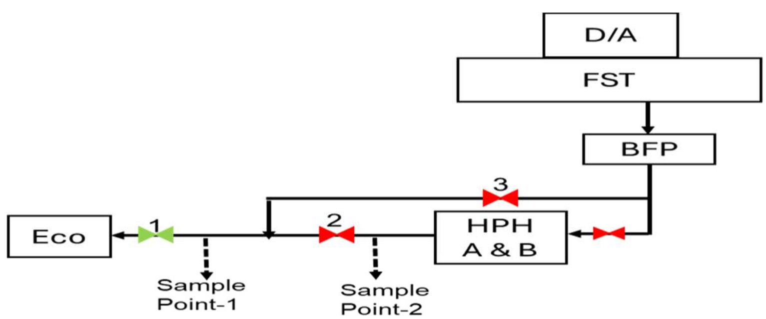
Like condensate system, HP heater circuit water recirculation is not possible due to non-availability of circulation line (except Khargone units). So, to avoid stagnation of water, HP heater circuit needs to be flushed on a regular interval *(minimum once/month)*. Flushing can be done by the following methods and depends on the boiler preservation type. For flushing, deaerator water quality should be as per ***Table- 5***.

**A. IF BOILER IS WET PRESERVED**

HP heater circuit water can be pushed to boiler for 10-15 minutes or more (depends on feed water flow) by boiler feed pump & from boiler, it can be drained to atmosphere.

**B. IF BOILER IS DRY PRESERVED**

HP heater circuit water can be flushed as per the scheme given in ***Fig.- 7.*** During HP heater circuit flushing, no water should be entered in economizer. This activity can be done with/ without start of BFP & flushing time can be decided accordingly. Water quality at the outlet of HP heater should be according to ***Table-5***.




| VALVES POSITION & SAMPLING POINT FOR HP HEATER FLUSHING | | | | | |
|---|---|---|---|---|---|
| Valve-1 Provided (at eco inlet ) | Valve-1 Position | Valve-2 Position (HP heater O/L v/v) | Valve-3 Position | Sample From Point-1 | Sample From Point-2 |
| Yes | Close | Open | Open/ Close | Yes | No |
| No | --- | Close | Close | No | Yes |

**Fig. 7:** *Schematic for Flushing (replacement) of HP Heaters*

After completion of LP & HP heater flushing/circulation, drain hot well and keep dry. The turbine dehumidifier should be in continuous service until OEM specified humidity value achieved inside turbine.

***Note:*** Any specific scheme or procedure recommended by the OEM for the preservation and periodic circulation of preservative water (if preserved wet) may be followed.

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### 9.2.3. PRESERVATION OF DEAERATOR

#### 9.2.a.1. SHUTDOWN DURATION < 4 DAYS

Deaerator temperature is to be maintained at 110 °C or more & pressure 2-3 kg/cm² by use of PRDS/auxiliary steam to avoid ingress of air in the deaerator. The preserved water (having pH ≥ 9.2) shall be used for refilling the boiler to get faster stabilization of Chemistry parameters during startup of unit.

If auxiliary steam source is not available, then keep deaerator in filled condition and maintain the same chemistry parameters as that of normal operation.

#### 9.2.a.2. SHUTDOWN DURATION > 4 DAYS

a) Increase pH in the deaerator during wet preservation of LP heaters (refer **Section-9.2.2.2**). Deaerator vents should be in closed condition.
b) An additional provision may be made available to purge and cap with nitrogen at de-aerator to preempt oxygen ingress.
c) Recirculate the FST water by boiler feed pump **fortnightly** and check & maintain the specified chemistry parameters (***Table-5***).
d) D/A water turbidity should be < 3 NTU.

***Note:***

i. *To prevent deaerator from corrosion, low pH DM water should not be stored in deaerator.*

ii. *The ammonia/hydrazine dosing lines can be extended to the deaerator filling line (without altering the existing dosing scheme for individual BFP suction) of the boiler feed pump, enabling direct dosing and proper mixing of chemicals in the deaerator water, particularly when DM makeup is added directly to the deaerator.*

### 9.2.b. PRESERVATION OF HOT WELL

Hot well shall be wet preserved up to 4 days if hot well pH is >9.0. After 4 days, drain the hot well and follow the dry preservation procedure as specified in ***Section- 9.1.4***.

Any specific guideline of layup provided by OEM can also be followed. A detailed protocol is to be filled up as per **Annexure-5**.

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### 9.3. NITROGEN GAS LAYUP PROECDURE

Nitrogen gas can be used for blanketing equipment that is drained but not completely dry, or for equipment filled with water to prevent air ingress. $N_2$ gas layup is similar to the instrument air preservation except that nitrogen is used at a slightly positive pressure on all components rather than instrument air purge. $N_2$ gas purity at each drain valve shall be checked. When $N_2$ gas purity becomes over 99% (or $O_2$ gas <1%) close all drain valves.

During preservation period, $N_2$ gas pressure at boiler inlet on any other equipment should be maintained at 0.3-0.5 Kg/cm$^2$. If the pressure drops below 0.3 kg/cm$^2$ recharge the pressure up to 0.5 Kg/cm$^2$. $N_2$ gas purity at each drain valve shall be checked at least once a week and record is to be maintained.

SAFETY:
Nitrogen does not support human life; therefore, relevant safety precautions are crucial. All nitrogen gas-filled accessible components must be properly labeled with a danger tag. In case of any maintenance need of such components, before personnel entry for maintenance, the system should be isolated from the nitrogen line and thoroughly purged with fresh air. The oxygen concentration should be at least 19.5% before personnel are allowed to enter

***Note:***

***i.*** Any specific changes in the cleanup and preservation scheme or chemistry parameter limits (better than this OGN), as recommended by the OEM, may also be incorporated into the station's LMI

***ii.*** A detailed protocol for the preservation of all equipment must be prepared as per **Annexure-3**, and it should be uploaded and maintained on the site intranet.

A summary of important chemistry parameters and their limits, to be monitored during the layup period, is provided in **Table 8**.

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
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**Table-8**: Summary of Chemistry Parameters & their Limits for Monitoring

| Component | Layup type | Parameter | Target value |
|---|---|---|---|
| Boiler & Superheater | Wet with nitrogen cap (If $N_2$ is not available, keep pressurized up to 10 $Kg/cm^2$) | pH | 10-10.5 (All ferrous) |
| | | | 9.5-10.0 (Mixed metallurgy) |
| | | Hydrazine | 200-250 ppm (Mixed metallurgy units) |
| | | Turbidity | <5 NTU |
| | Dry with inst. air | Relative humidity of purge-out air | < 35% |
| | Dry with $N_2$ gas | $O_2$ % at drain point | < 1% |
| Reheater | Dry with inst. air | Relative humidity of purge-out air | < 35% |
| | Dry with $N_2$ gas | $O_2$ % at drain point | < 1% |
| Heaters (Tube side) | Wet preservation | pH | 10-10.5 (all ferrous units) |
| | | | 9.0 -9.2 (Mixed metallurgy) |
| | | Hydrazine | 40-50 ppm (Mixed metallurgy units) |
| | | Turbidity | < 3NTU |
| | Dry with inst. air | Relative humidity of purge-out air | < 35% |
| Heaters | Dry with inst. air | Relative humidity | < 35% |

**Table-9**: Summary of Chemistry Parameters & their Limits for Monitoring

| Component | Layup type | Parameter | Target value |
|---|---|---|---|
| (shell side) | Dry with $N_2$ gas | $O_2$ % at drain point | < 1% |
| | Wet in hot condition | Feed water storage tank temperature | > 110 $^0C$ |

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| | | | |
|---|---|---|---|
| Deaerator | | pH | ≥ 9.2 (8.8-9.3 for mixed metallurgy units) |
| | Wet in cold condition with or without $N_2$ cap | pH | 10-10.5 (all ferrous units) |
| | | | 9.0 -9.2 (Mixed metallurgy) |
| | | Hydrazine | 40-50 ppm (Mixed metallurgy units) |
| | | Turbidity | < 3NTU |
| | Dry with inst. air | Relative humidity of purge-out air | < 35% |
| Hot well | Drained and keep dry | RH | < 35 % |
| | Wet for short duration | pH | ≥ 9.2 (8.8-9.3 for mixed metallurgy units) |

### 9.4. TRANSITIONING FROM WET LAYUP TO DRY LAYUP

There are occasions when a steam cycle has been preserved in wet, and it is desirable or necessary to transition to a dry layup condition.

**a)** Transitioning from wet layup to dry layup is similar to transitioning from normal operation to dry layup. Presuming that the boiler was properly laid-up (as specified in ***Section-9.2.1***), the boiler can be heated by boiler light up to achieve SH achromat temperature > 110 $^0$C and SAPH flue gas inlet temperature > 220 $^0$C and water wall temperature > 110 $^0$C.

**b)** For boiler hot draining and air filling, follow the steps as given in ***Section-9.1.1.2***. **Ensure that water in SH and reheater coils must evaporate** before the start of instrument air injection (Acromat temperature > 110 $^0$C and SAPH flue gas inlet temperature > 110 $^0$C and no steam vapour coming from the SH vent)

**c)** For boiler dry preservation with nitrogen gas, refer to *Section-9.*3.

**d)** It is to be noted that high pH water should be replaced with normal operating pH water before boiler light-up.

**e)** For drum-type units, boiler water turbidity should be <2 NTU before light-up of the boiler.

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## 9.5. PRESERVATION OF AUXILIARY BOILERS

**a)** Auxiliary boilers can be preserved similar to main boiler (i.e. Dry with Instrument air/ Dry with $N_2$ gas, wet with $N_2$ capping or Wet pressurized). Periodic checking of condition of auxiliary boiler is to be done and recorded.

**b)** For faster startup, if the metallurgy of the boiler and auxiliaries is all ferrous and is **completely drainable,** then it can be preserved dry with instrument air/dehumidified air in line with the main boiler (Refer ***to Section 9.1***).

**c)** A station-specific schematic drawing, based on guidelines (Refer ***to Section 9.1.1.2***) & local conditions, shall be prepared by the site and may be vetted by the COS-Boiler group before its implementation.

**d)** If the auxiliary boiler is wet preserved, it should be completely drained and filled with fresh treated DM water before lighting up.

**e)** If the main boiler feed water regime is AVT (O)/OT, then hydrazine is not to be used in the auxiliary boiler during their layup & normal operation.

**f)** A detailed protocol is to be filled up as per ***Annexure-3***.

## 9.6. HYDRO TESTS OF BOILER

**a)** The temperature of water used as medium of pressure testing shall not be less than 20 $^0$C and more than 50 $^0$C (IBR regulations-379).

**b)** For hydro test, boiler can be filled directly from boiler fill pump or deaerator water. Deaerator flushing is required for this purpose.

**c)** If the unit is returning after a boiler overhaul or modification, the superheater should be filled through the backfilling line only, to avoid the ingress of rust and debris from the boiler water wall tubes into the superheater.

**d)** For hydro test of normal operating units, the superheater can be filled by overflowing the boiler water after ensuring separator/boiler water parameters as specified in **Table 10.**

**e)** For boiler hydro test, treated DM water of the following quality (as specified in ***Table 10***) shall be used when the boiler light-up is to be done within 4 days after the completion of the boiler hydro test.

**f)** High concentration of chemicals (***Table 9***) is to be used when boiler light-up is planned for later. After hydro test, keep the boiler under pressure conditions of ~10 kg/cm² for wet preservation.

**g)** After completion of hydro test, boiler must be drained before proceeding with unit startup.

**h)** Unit start up to be done as per procedure specified in **Section 7.0/ 8.0**.

| **Table-10:** Filling Water Quality for Boiler hydro test | | |
|---|---|---|
| Parameters | FW Regime-AVT(R) | FW Regime-AVT(O)/OT |
| Hydrazine, ppb | 20-30 | -- |
| pH at 25 $^0$C | 9.0- 9.2 | 9.2- 9.6 |
| Turbidity (NTU) | <1.0 | <1.0 |

| **Table-11:** Filling Water Quality for Boiler hydro test | | |
|---|---|---|
| Parameters | FW Regime-AVT(R) | FW Regime-AVT(O) |
| Hydrazine, ppm | 200-250 | -- |
| pH at 25 $^0$C | 9.5-10 | 10-10.5 |
| Turbidity (NTU) | <1.0 | <1.0 |

10. LAYUP PROCEDURE FOR TURBINE

a. Corrosive attack of turbine components during long shutdown periods typically occurs by the interaction of air (oxygen) and moisture with impurities that were formed during operation. A small amount of salt accumulation on a turbine blade, for example, can result in severe corrosion like **SCC** when the unit is shut down.

b. Dry layup of turbines shall be done by dehumidified air as per procedure specified by OEM. An example of a dry layup scheme of a steam turbine is given below (***Fig. 8***).

c. Relative humidity should be monitored frequently to ensure that dehumidification is adequate to protect turbine components. A detailed protocol is to be filled up as per ***Annexure-3***.

d. The LP turbine set may require a continuous dry airflow to prevent moisture accumulation from the condenser/hot well if not drained and dried.

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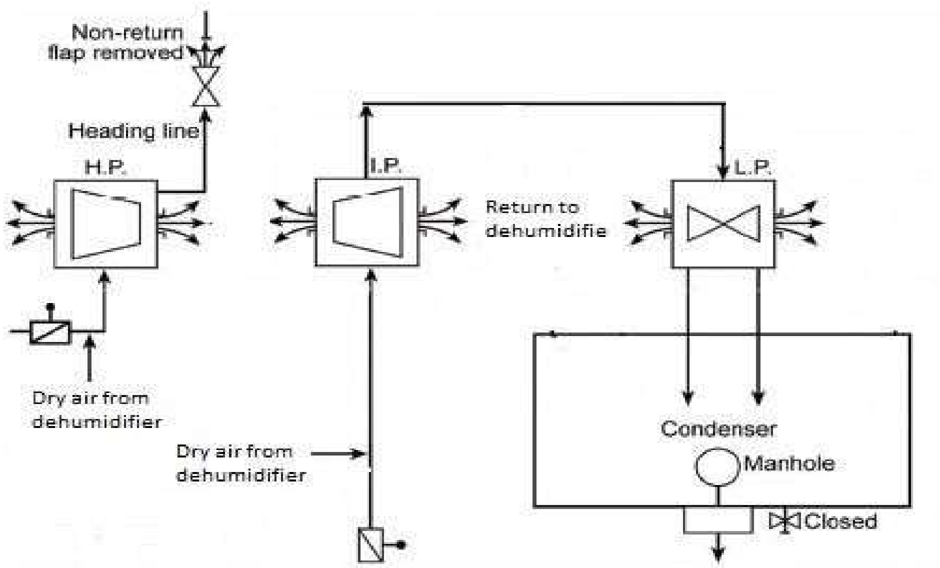
**Figure-8**: Schematic for dry layup of turbine using dehumidified air




## 11. THE SAMPLING OF STEAM AND WATER FOR CHEMICAL MONITORING AND ANALYSIS

It is essential that the SWAS sample panel be in service as soon as possible during the startup process to provide reliable and representative samples for grab sampling and instrument supply. It is also essential that the core monitoring instruments be in service, providing reliable data as soon as possible during the unit startup process.

In power plants, generally two types of common sampling facilities are provided: one is a local sampling rack, which is required for grab sampling during startup, if required, and another is for SWAS (Steam Water Analysis System), which consists of a primary rack and a secondary rack (wet panel).

### 11.1 SWAS SAMPLES DETAILS

#### 11.1.1. BOILER WATER SAMPLE

- The fluid is typically at high temperature and pressure. This sample line, provided in drum-type units, is used to monitor the salt content in the boiler water and to facilitate blowdown and control measures.
- To obtain a representative boiler water sample in 500 MW units, it should be extracted from the down comer line of the boiler, i.e., from the common suction header of the boiler circulation pumps (BCP).

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#### 11.1.2 STEAM SAMPLES

- Two types are used for sampling of steam: one is saturated steam to monitor salt carryover and steam dryness, and another is the main steam sample to monitor the quality of steam entering the turbine.

#### 11.1.3 CONDENSATE AND CPU SAMPLES

- This sample point is provided from the CEP common discharge header. This sample is used to monitor the ingress of any contaminants in the hot well from DM makeup, condenser cooling water, air ingress, etc.
- CPU sample line is also provided from the condensate discharge header but downstream of the CPU. To get a representative sample of CPU in SWAS, it should be extracted from upstream of the LP chemicals (ammonia, hydrazine & oxygen) dosing points (the gap should be a minimum of 2-3 feet) at CEP & CPU common discharge header.
- For units equipped with CPU, the ammonia dosing line should be provided after GSC recirculation line to avoid early exhaustion of service-in CPUs during start-up.
- Refer to **Annexure-8** for standard schematics for condensate dumping, dosing, and sampling systems.

#### 11.1.4 DEAERATOR SAMPLE

a) For mixed metallurgy unit, deaerator inlet sample is provided for monitoring of ORP and dissolved oxygen.

b) Deaerator outlet sample is taken from the suction of each MDBFP and TDBFP. This sample is used to monitor the performance of the deaerator. This is also used for monitoring of dissolved oxygen at deaerator outlet to control/optimize the deaerator venting. The sample line should be provided from upstream of the LP chemicals (ammonia, hydrazine & oxygen) dosing points.

c) For measurement of deaerator DO during startup, one local sample point with an isolation valve can be provided from the cooler discharge of deaerator sample at primary rack. Refer to Annexure-7 for the schematic.

#### 11.1.5 FEED WATER SAMPLE

This sample point is extracted from the economizer inlet header. Feed water sample line should not be extracted from the bottom of the economizer inlet line. Feed water sample is used to monitor and control feed water quality to the boiler.

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## 11.2 SAMPLE TRANSPORT SYSTEM

**a)** Samples are transported in suitable material like SS-304, 316, or similar metallurgy that does not contaminate samples in any way. Sample line made of MOC: MS or CS should not be used for the transport of any samples.

**b)** The entire sample tube length should be adequately thermally insulated and labelled.

**c)** For safe and reliable sampling, double isolation valves (root valves) should be provided for all high-pressure steam and water samples, such as feed, boiler, steam, etc.

**d)** Blowdown lines should be available for each sample in the local sampling rack as well as in primary and secondary sample racks.

**e)** Stations experiencing low sample flow during partial load operation or startup may install an automatic pressure-reducing system at the primary sample rack to get consistent flow in the sample line.

## 11.3 SAMPLE CONDITIONING SYSTEM

**a)** Temperature reduction should always be carried out before pressure reduction to prevent flashing, which results in deposit formation.

**b)** Primary coolers are used to cool the sample to approximately 38-45°C.

**c)** The secondary coolers are used to cool the sample, and maintain it at 25°±0.5°C. This eliminates the necessity of making temperature corrections for pH and conductivity, thus improving the accuracy of these measurements.

**d)** The mechanical (local) sample coolers can be used in series to primary cooler for better cooling of steam samples. This modification can be done during unit overhauling. Refer to **Annexure-6** for schematic.

**e)** There should be two different power sources for compressors and pumps of the SWAS chiller system. Also, the chiller system should have a cooling water makeup source from two different units. Both will ensure uninterrupted availability of the SWAS chiller system.

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## 12 RECORDS TO BE MAINTAINED.

| No. | Record description | Maintained by | Frequency | Remarks |
|---|---|---|---|---|
| 1 | Signed protocols | Chemistry | As specified in different annexures | |
| 2 | Signed templates | Chemistry | As specified in different annexures | |

## 13 DISTRIBUTION OF LMI

A soft copy of the LMI shall be uploaded in the EEMG dept. website on Simhadri intranet LAN by the EEMG dept. (Password Protected) The uploaded soft copy of the document is to be considered as Controlled Copy.

## 14 REVISION OF LMI

Head of Chemistry is responsible for put up approval from Competent Authority and for revision of LMI the document on a 2-yearly basis or as and when required.

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**ANNEXURE-1A: UNIT START UP CHECK LIST/ PROTOCOL FORMAT FOR SUBCRITICAL UNITS (DRUM TYPE UNITS)**

Station: Unit:

Shutdown Reason: Shutdown Duration:

| Sl. No | Equipment/ System | Status |
|---|---|---|
| 1 | Ammonia dosing pumps status and tanks level | |
| 2 | Hydrazine dosing p/ps status and tanks level | |
| 3 | HP dosing pumps status and tanks level | |
| 4 | CPU service vessel-A & B availability status | |
| 5 | CPU service vessel-A & B totalizer reading($M^3$) | |
| 6 | SWAS Chiller Status | |
| 7 | SWAS Analysers Status (Availability & Calibration) | |

<table>
<tr><td>Date/ Activity Start & End Time (hrs)</td><td>Activity</td><td>Parameters<br>(Refer to Section-7.0)</td><td colspan="4">Limiting Values<br>(Refer to Section-7.0)</td><td>Clearance value</td></tr>
<tr><td rowspan="3">Date:<br>Start time:<br>End Time:</td><td rowspan="3">Hot well & CEP discharge line clean up</td><td colspan="5">Sample name: Condensate dumping water</td><td></td></tr>
<tr><td>Silica (ppb), max</td><td colspan="4">50 (100 for units with CPU)</td><td></td></tr>
<tr><td>Turbidity (NTU), max</td><td colspan="4">2 (5 for units with CPU)</td><td></td></tr>
<tr><td rowspan="9">Date:<br>Start time:<br>End Time:</td><td rowspan="9">LP Heaters and Deaerator open flushing</td><td>Sample name: FST drain (deaerator outlet)</td><td>Mixed met. w/o CPU</td><td>Mixed met. with CPU</td><td>All ferrous (210/ 250 MW)</td><td>All ferrous (500 MW)</td><td></td></tr>
<tr><td>pH at 25 $^0$C</td><td>8.8-9.3</td><td>8.8-9.3</td><td>9.2-9.6</td><td><9.6</td><td></td></tr>
<tr><td>Silica (ppb), max</td><td>50</td><td>30</td><td>50</td><td>100</td><td></td></tr>
<tr><td>Turbidity (NTU), max</td><td>1.5</td><td>1.0</td><td>1.5</td><td>5.0</td><td></td></tr>
<tr><td>CRUD (ppb), max</td><td>---</td><td>50</td><td>--</td><td>--</td><td></td></tr>
<tr><td>Total iron (ppb), max</td><td>100</td><td>50</td><td>100</td><td>--</td><td></td></tr>
<tr><td>Residual hydrazine (ppb)</td><td>20-30</td><td>20-30</td><td>--</td><td>--</td><td></td></tr>
<tr><td>Total Copper, ppb, max</td><td>10</td><td>10</td><td>--</td><td>--</td><td></td></tr>
<tr><td colspan="5">Sample name: CEP discharge</td><td></td></tr>
</table>

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

| | Activity | Parameters (Refer to *Section-7.0*) | Limiting Values (Refer to *Section-7.0*) | | | | Clearance value |
|---|---|---|---|---|---|---|---|
| | | CC/DCC at 25 $^0$C (µS/cm), max | 1.0 | 1.0 | 1.0 | -- | |
| Date: Start time: End Time: | LP Heaters and Deaerator closed circulation | **Sample name:** FST drain | | | | | |
| | | pH at 25 $^0$C | --- | --- | --- | 9.2-9.6 | |
| | | Silica (ppb), max | --- | --- | --- | 30 | |
| | | Turbidity (NTU), max | --- | --- | --- | 1.0 | |
| | | CRUD (ppb), max | --- | --- | --- | 50 | |
| | | Total iron (ppb), max | --- | --- | --- | 50 | |
| | | **Sample name:** CEP discharge | | | | | |
| | | CC/DCC at 25 $^0$C (µS/cm), max | --- | --- | --- | 1.0 | |
| Date Start End: | FST heating | **Sample name:** FST/ deaerator outlet | | | | | |
| | | FST temperature ($^0$C), min. | 105 | | | | |
| | | D.O. (ppb), max | 100 | | | | |
| Date: | HP heaters & FW line flushing | Flushing started at: Flushing completed at: | | | | | |
| | **Boiler light up date & time:** | | | | | | |
| Date: | Boiler hot Cleanup & Steam dumping and turbine rolling clearance | **Sample name:** Main Steam (for turbine rolling) | | | | | |
| | | pH at 25 $^0$C | 8.8-9.3 | 8.8-9.3 | 9.2-9.6 | 9.2-9.6 | |
| | | Silica (ppb), max | 20 | 20 | 20 | 20 | |
| | | Sodium (ppb), max | 5 | 5 | 5 | 5 | |
| | | CC at 25 $^0$C (µS/cm), max | -- | -- | -- | 1.0 | |
| | | Total iron (ppb), max, | 10 | 10 | 10 | 10 | |
| | | CRUD (ppb), max | Nil | Nil | Nil | Nil | |
| | | **Sample name:** CEP discharge | | | | | |
| | | CC/DCC at 25 $^0$C (µS/cm), max | 1.0 | 1.0 | 1.0 | 1.0 | |

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

| Sample name: CBD Sample | | | | | |
|---|---|---|---|---|---|
| Drum pressure (Kg/cm$^2$) | -- | -- | -- | -- | |
| pH at 25 $^0$C | 9.2-9.6 | 9.2-9.6 | 9.2-9.6 | 9.2-9.6 | |
| Sp. cond at 25 $^0$C (µS/cm), PT | 20 | 20 | 20 | 15 | |
| Phosphate (ppm), Lab value | 0.2-2.0 | 0.2-2.0 | 0.2-2.0 | 0.2-2.0 | |
| Chloride (ppb), max, Lab value | 500 | 500 | 500 | 200 | |

| Date/ Activity Start & End Time (hrs) | Activity | Parameters (Refer to *Section-7.0*) | | | | Limiting Values (Refer to *Section-7.0*) | | | | Clearance value |
|---|---|---|---|---|---|---|---|---|---|---|
| | **Turbine rolling clearance date & time:** | | | | | | | | | |
| | **Unit Synchronization date & time:** | | | | | | | | | |
| | HP Heaters Drip line flushing | Flushing duration:<br>HP heaters drip line charged to deaerator at (hr): | | | | | | | | |
| | DPR* details | Time | Unit Load | Drum pressure | CBD open Status | CBD SiO2 | MS SiO2 | DPR* | Condensate flow | CPU-A Flow | CPU-B Flow |
| | | Hrs. | MW | kg/cm$^2$ | % | ppb | ppb | kg/cm$^2$ | M$^3$/hr | M$^3$/hr | M$^3$/hr |
| | *: Drum pressure restriction. | | | | | | | | | | |
| | | | | | | | | | | | |
| Date: | | | | | | | | | | | |
| | | | | | | | | | | | |
| | **SWAS samples charging details:** | | | | | | | | | |
| | **Deviation in the startup process with respect to guidelines:** | | | | | | | | | |

**Name & Signature:**

**(Chemistry)** **(Operation)**

***Note:*** *The signed copy of the startup protocol must be uploaded and maintained on the site intranet by the chemistry team.*

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## ANNEXURE-1B: TEMPLATE FOR RECORDING STARTUP CHEMISTRY PARAMETERS FOR SUBCRITICAL UNITS (DRUM TYPE UNITS)

| **Sample** | **Unit Load** | **Drum Pressure** | **CEP Discharge** | | **CPU-A** | | | **CPU-B** | | | **D/A O/L** | **Feed Water** | | | **Boiler Water** | | | | | **Main Steam** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Silica | Turb. | Flow | Silica | Sp. Cond | Flow | Silica | Sp. Cond | Silica | Sp. Cond | Silica | Turb. | pH | Sp. Cond | Silica | Phosp | Turb. | Silica | Turb. |
| Date & time | MW | kg/cm$^2$ | ppb | NTU | M$^3$/hr | ppb | µS/cm | M$^3$/hr | ppb | µS/cm | ppb | µS/cm | ppb | NTU | -- | µS/cm | ppb | ppb | NTU | ppb | NTU |
| | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | |

**Note:** All parameters must be recorded from unit startup until the stabilization of chemistry parameters at full-load operation. The sample analysis frequency should be once every two hours. However, if DPR is imposed due to high steam silica; the Feed water, Boiler water, and Main steam samples should be analyzed at 30-minute intervals

Name & Signature: (Chemistry)

ANNEXURE-2:
**SCHEMATIC DRAWING FOR DRY AIR PRESERVATION OF DRUM TYPE UNITS**

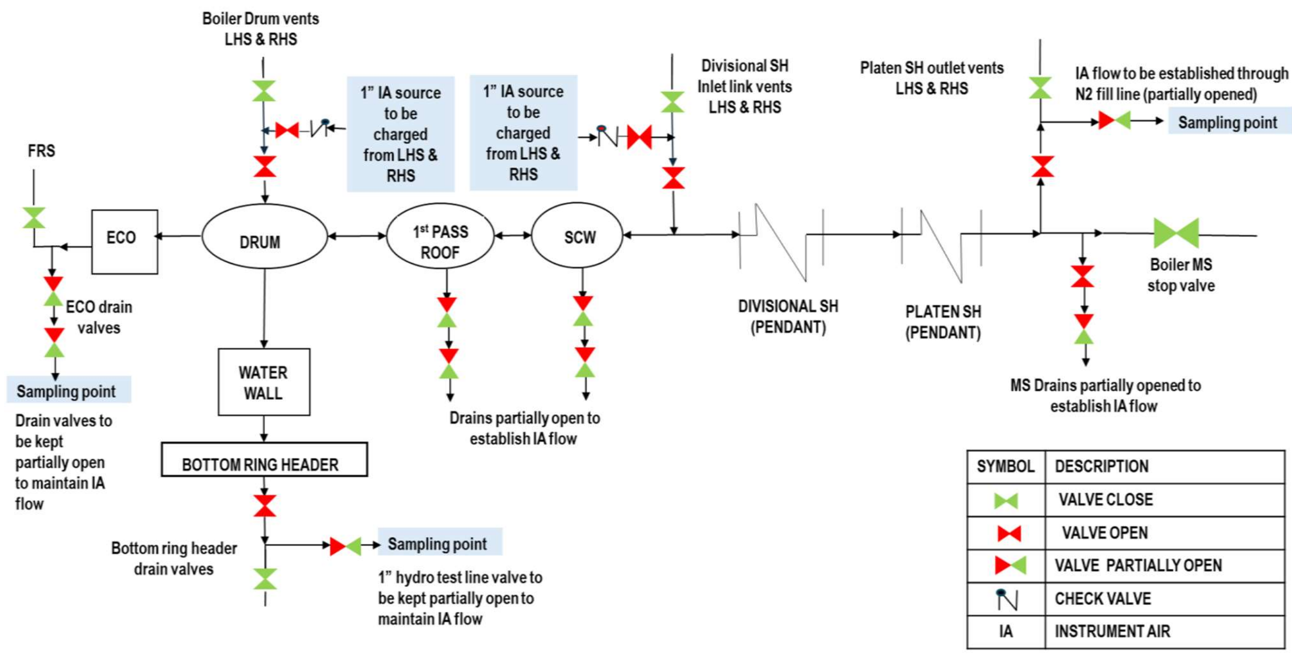



**Fig-9:** Schematic Drawing for Dry air Preservation of Drum type Boiler

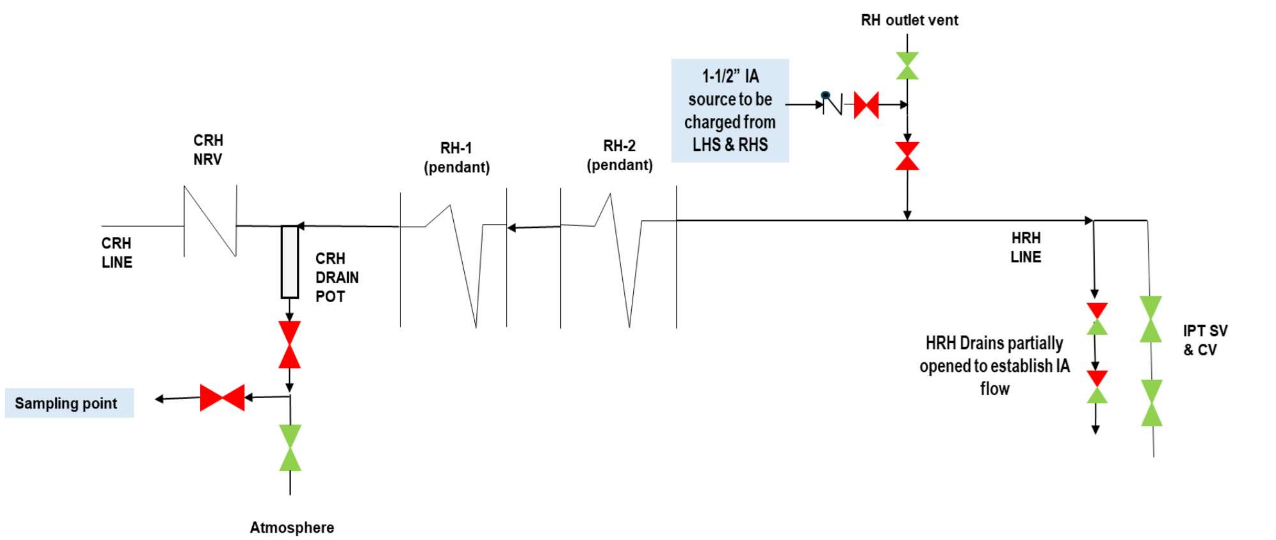



**Fig-10:** Schematic Drawing for Dry air Preservation of Reheater circuit of Drum type Boiler

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## ANNEXURE-3: PROTOCOL ON PRESERVATION STATUS

(Refer ***Section-11.0, 12.0 & 13.0***)

STATION NAME :

UNIT : DATE & TIME:

UNIT SHUTDOWN DATE: PRESERVATION START DATE:

TYPE OF LAYUP : Dry with instrument air/ Dry with $N_2$ / Wet with $N_2$ cap/ Wet pressurized

**A. NAME OF SYSTEM/EQUIPMENT KEPT UNDER PRESERVATION:**

**B. IF UNDER DRY PRESERVATION:**

Dew point ($^0$C) of Instrument Air/ RH (%) of dry air (inlet) :

Relative Humidity (%) of dry air (outlet) :

Dehumidifier Inlet/ Out temperature ($^0$C) :

Air/ $N_2$ pressure inside preserved equipment (Kg/cm$^2$) :

**C: IF UNDER WET PRESERVATION:**

BCP running time for circulation (hour) :

Boiler pressure (Kg/cm$^2$) :

$N_2$ pressure (Kg/cm$^2$) :

**CHEMICAL PARAMETERS:**

pH :

Sp. Cond. (uS/cm) :

Hydrazine (ppm) :

Turbidity (NTU) :

CRUD* (ppb) :

Chloride** (ppb) :

**D: Quantity of makeup chemicals if any (Liter) :**

**Note:**

*i.* During the initial filling of the system, pH and hydrazine levels in the vent samples of each system (boiler, superheater, etc.) must be checked and recorded.

*ii.* When the boiler is under wet preservation, the boiler water pH must be checked after recirculation by the BCP. For units without a BCP, refer to **Section 11.2.1.2-f**.

*iii.* *: For boiler water sample only.

*iv.* **: For boiler water of drum type boilers only.

*v.* Refer to **Table-18** for the limits of pH, hydrazine, and turbidity. Specific conductivity and CRUD should be analyzed to maintain records and compare results over time.

*vi.* Keep a record of CRUD papers.

The Chloride and phosphate should be BDL in boiler water sample

**NAME & SIGNATURE:**

**(Chemistry)** **(Operation)**

***Note:*** *The signed copy of preservation protocol to be uploaded & maintained at site intranet by chemistry*

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## . ANNEXURE-4: CRUD TESTING PROCEDURE (COLOUR COMPARISON METHOD)

### SUMMARY OF METHOD

The sample is filtered through a membrane filter of uniform porosity of 0.45 micron and the coloration produced on the membrane filter is compared with the standard Babcock & Wilcox Comparison Chart.

### APPARATUS

i. Filter Holder, 47mm diameter
ii. Filtration Flask, 1 litre capacity.
iii. Laboratory Vacuum pump.
iv. Cellulose nitrate membrane filter paper, 47mm diameter, 0.45-micron porosity.
v. Stainless steel frit (filter screen) with uniform porosity (preferably)
vi. Hot air oven to dry the filter paper.

### PROCEDURE

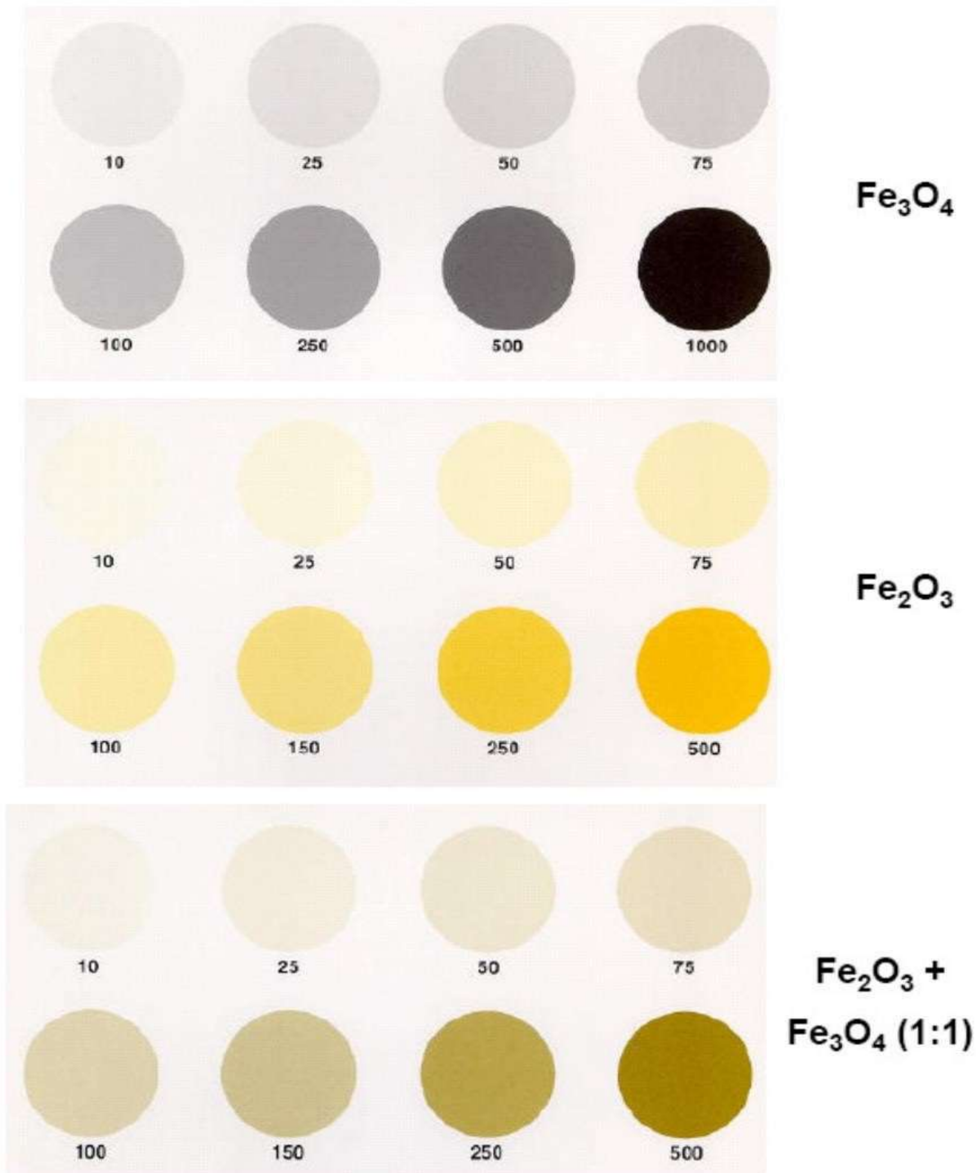
Filter 1-litre of the sample through the membrane filter by applying vacuum. Dry the membrane filter at room temperature or hot air oven and compare the coloration on the membrane filter with the standard Babcock & Wilcox Comparison Chart.

### CALCULATIONS

Determine the CRUD level in the sample by matching the color with the enclosed Babcock & Wilcox Comparison Chart. This method provides approximation only. To determine the exact CRUD level, the weighing method should be adopted after passing an adequate amount of water through the membrane using an in-line filter holder.

### NOTE:

Keep records of CRUD papers used in startup activities for clearance. Details such as the unit, sample name, date and time, name of the startup activity, CRUD value (by color comparison), etc., should also be noted for future reference.




The color and density of the stains on this chart are approximately the same as those produced when one liter of water containing the indicating concentration of suspended Iron in parts per billion (Fe) is passed through white filter membrane having a pore size of 0.45 microns.

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## ANNEXURE-5:
## STANDARD PI TAG LIST FOR MONITORING OF SWAS & CPU PARAMETERS DURING UNIT START UP & NORMAL OPERATION

| PI TAG LIST FOR DRUM TYPE UNITS (WITH CPU) | | | |
|---|---|---|---|
| **S.N.** | **Parameters** | **S.N.** | **Parameters** |
| 1 | DM makeup water Sp. Conductivity | 31 | Main steam Sp. Conductivity |
| 2 | DM makeup water CC | 32 | CPU-A flow |
| 3 | DM makeup water DCC | 33 | CPU-A CC |
| 4 | CEP discharge CC | 34 | CPU-A Sp. Conductivity |
| 5 | CEP discharge DCC | 35 | CPU-A Sodium |
| 6 | CEP discharge D. O. | 36 | CPU-A Vessel DP |
| 7 | CEP discharge Sodium | 37 | CPU-A RT DP |
| 8 | CEP discharge Sp. Conductivity | 38 | CPU-A vessel pressure |
| 9 | CEP discharge pH | 39 | CPU-B flow |
| 10 | CPU o/l Sp. Conductivity (SWAS) | 40 | CPU-B CC |
| 11 | CPU outlet CC (SWAS) | 41 | CPU-B Sp. Conductivity |
| 12 | CPU outlet Sodium (SWAS) | 42 | CPU-B Sodium |
| 13 | CPU outlet pH (SWAS) | 43 | CPU-B Vessel DP |
| 14 | D/A outlet D.O. | 44 | CPU-B RT DP |
| 15 | Feed water pH | 45 | CPU-B vessel pressure |
| 16 | Feed water Sp. conductivity | 46 | CPU common DP |
| 17 | Feed water CC | 47 | CPU bypass valve position |
| 18 | Feed water DO | 48 | Hot well Conductivity (Left) |
| 19 | Boiler water pH | 49 | Hot well Conductivity (Right) |
| 20 | Boiler water Sp. conductivity | 50 | SWAS Chiller Temperature |
| 21 | Boiler water CC | 51 | CCW pH |
| 22 | Boiler water Chloride | 52 | CCW Sp. Conductivity |
| 23 | Boiler water Silica | 53 | CCW Turbidity |
| 24 | Boiler water Phosphate | 54 | CCW ORP |
| 25 | Saturated steam CC | 55 | Condensate flow to deaerator |
| 26 | Saturated steam Sodium | 56 | Ammonia metering tank level |
| 27 | Saturate steam Sp. Conductivity | 57 | Amm. dosing p/p dich header pressure |
| 28 | Main steam pH | 58 | Hydrazine metering tank level |
| 29 | Main steam CC | 59 | Hydzn. dosing p/p dich header pressure |
| 30 | Main steam Sodium | 60 | Hot well temperature |

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## ANNEXURE-6:
## SCHEMATIC FOR SWAS PRIMARY RACK SAMPLE COOLER

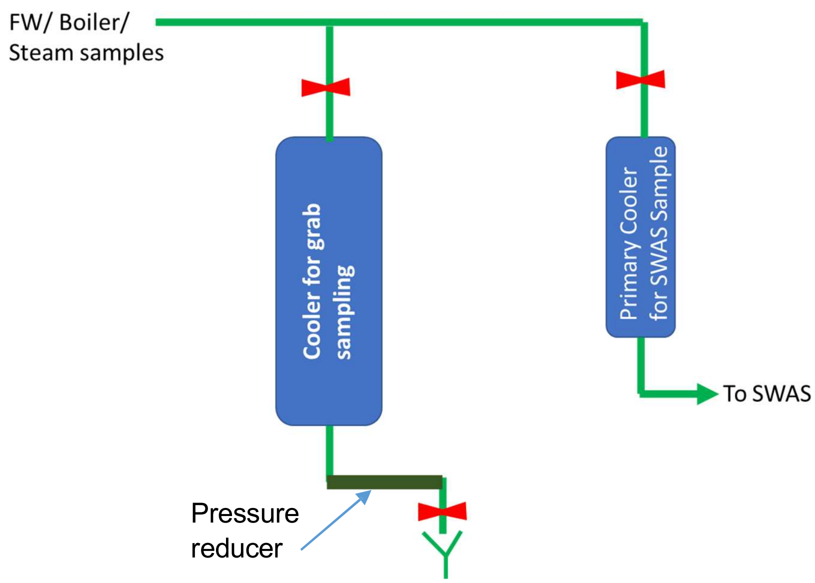



**Fig-11:** SWAS PRIMARY COOLER-ORIGINAL SCHEME

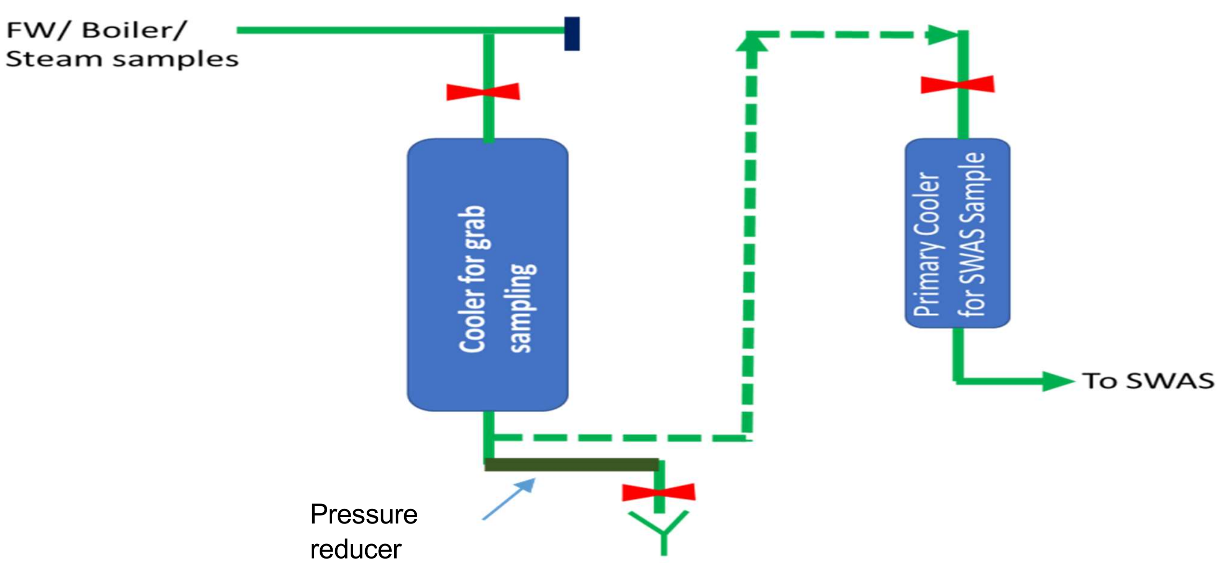



**Fig-12:** SWAS PRIMARY COOLER-MODIFIED SCHEME

| TITLE | CYCLE CHEMISTRY GUIDELINES FOR THE START UP, SHUT DOWN AND LAYUP PRACTICES |
|---|---|
| LMI No. | LMI/OGN/OPS/SYST/065, REV:0 |

## ANNEXURE-7: SCHEMATIC FOR DEAERATOR LOCAL SAMPLE

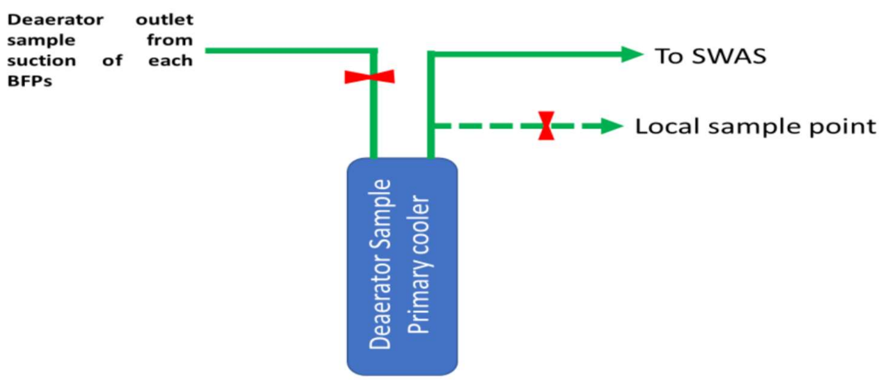
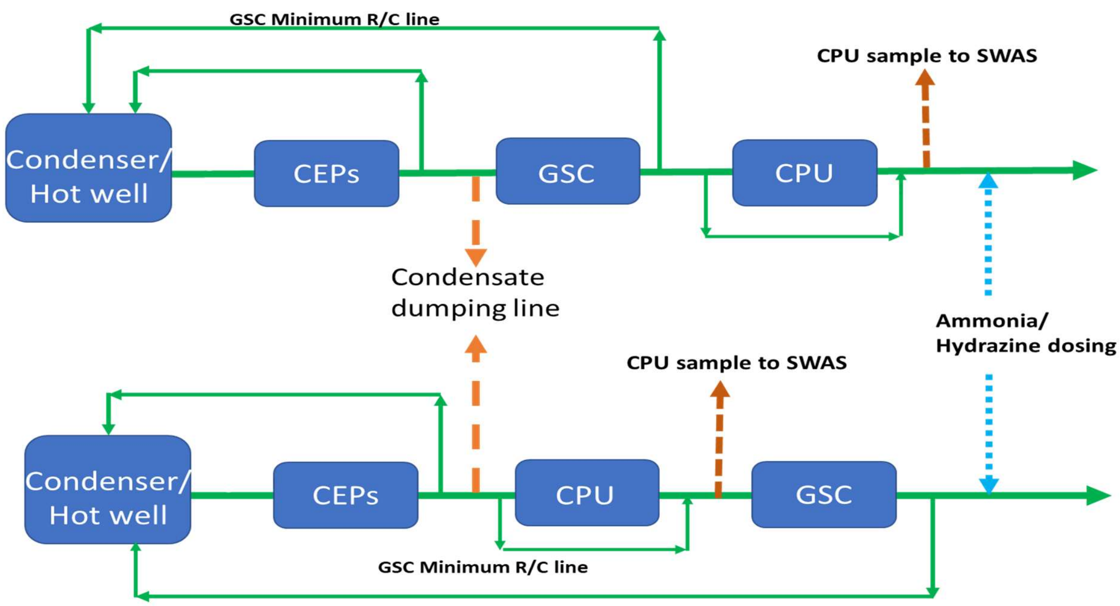



Fig-13: **Schematic for Deaerator Outlet Local Sample**

## ANNEXURE-8: SCHEMATIC FOR CONDENSATE DUMPING, DOSING, AND SAMPLING SYSTEM




**Note:** The size of condensate dumping pipelines should be 100-200NB for 500/660/800MW units and 80-100NB for 110/210/250MW units.